Notes

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The bad news is that we have still a long way to go.
The good news is that we have still a long way to go.

— Alain Fournier, 1998

Contents

[1] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

1. Tori, Pinnings, and Coherent Continuation

2023-12-22

1.1. **Identification of co-character groups with Lie algebras.** Settings: G complex connected reductive algebraic group. $Q \subseteq P \subseteq \mathfrak{h}^*$ are root lattice and weight group, respectively. $\check{Q} \subseteq \check{P} \subseteq \mathfrak{h}$ are coroot lattice and coweight group, respectively. H is a maximal torus of G, $\mathfrak{h}$ is its Lie algebra.

There are identifications

$$\begin{array}{rccc} I\colon & X_*(H) \otimes \mathbb{C} & \longrightarrow & \mathfrak{h} \\ & \phi & \longmapsto & \mathrm{d}\phi(1) \end{array}$$

$$\begin{array}{rccc} J\colon & X^*(H) \otimes \mathbb{C} & \longrightarrow & \mathfrak{h}^* \\ & \varphi & \longmapsto & \mathrm{d}\varphi \end{array}$$

Under these identifications we have

$$\begin{aligned} P_* &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \langle \alpha, \check{\lambda} \rangle \in \mathbb{Z}, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \exp(2\pi\sqrt{-1}\, \langle \alpha, \check{\lambda} \rangle) = 1, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \alpha(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \alpha \in Q \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \exp(2\pi\sqrt{-1}\check{\lambda}) \in Z(G) \right\}. \end{aligned}$$

And

$$\begin{aligned} X_*(H) &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \langle \phi, \check{\lambda} \rangle \in \mathbb{Z}, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \exp(2\pi\sqrt{-1}\, \langle \phi, \check{\lambda} \rangle) = 1, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \phi(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \phi \in X^*(H) \right\} \\ &= \left\{ \check{\lambda} \in \mathfrak{h} \;\middle|\; \exp(2\pi\sqrt{-1}\check{\lambda}) = 1 \right\}. \end{aligned}$$

Similarly

$$P^* = \left\{ \lambda \in \mathfrak{h}^* \;\middle|\; \exp(2\pi\sqrt{-1}\lambda) \in Z(\check{G}) \right\}.$$

$$X^*(H) = \left\{ \lambda \in \mathfrak{h}^* \;\middle|\; \exp(2\pi\sqrt{-1}\lambda) = 1 \right\}.$$

Define $e : \mathfrak{h} \to H$ by $e(X) = \exp(2\pi\sqrt{-1}X)$, then $\ker(e) = X_*(H)$.

$$\begin{tikzcd} \mathfrak{h} \arrow[r, "e"] & H \\ P_* \arrow[u, hook] \arrow[r, "e"] & Z(G) \arrow[u, hook] \end{tikzcd}$$

$Z(G) \simeq P_*/X_*(H)$.

1.2. **Pinnings of algebraic groups.**

Theorem 1.1 (see [AC09])**.** *Inner automorphism group* $\mathrm{Inn}(G)$ *of* G *is equal to* G/Z(G), $\mathrm{Inn}(G)$ *acts freely and transitively on the set of pinnings.*

I know that there is an analogy in compact connected groups, and it has been proved.

Theorem 1.2. *If* G *is a compact connected group, then its inner automorphism group* $\mathrm{Inn}(G) =$ G/Z(G) *acts freely and transitively on the set of pinnings. (where pinnings for it is a bit different from the complex case, that* $(T, B_{\mathbb{C}}, \mathcal{X})$ *is a pinning if* T *is a maximal torus,* $B_{\mathbb{C}}$ *a Borel of* $G_{\mathbb{C}}$ *containing* T*,* $\mathcal{X}$ *is a set of real rays in simple root space.)*

Proof. Here's proof from mathoverflow LSpice's answer:

To prove this, I'll use a few pieces of structure theory:

1. All maximal tori in G are G°-conjugate.
2. All Borel subgroups of $G_{\mathbb{C}}$ are $G^\circ_{\mathbb{C}}$-conjugate.
3. For every maximal torus T in G, the map $\mathrm{W}(G^\circ, T) \to \mathrm{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ is an isomorphism.
4. If G_{sc} and $(G_{\mathbb{C}})_{\mathrm{sc}}$ are the simply connected covers of the derived groups of G° and $G^\circ_{\mathbb{C}}$, then $(G_{\mathrm{sc}})_{\mathbb{C}}$ equals $(G_{\mathbb{C}})_{\mathrm{sc}}$.
5. Every compact Lie group has a finite subgroup that meets every component.

I only need (4) to prove that, for every maximal torus T in G, the map from T to the set of conjugation-fixed elements of $T/\mathrm{Z}(G^\circ)$ is surjective. This is probably a well known fact in its own right for real-group theorists.

Now consider triples $(T, B_{\mathbb{C}}, \mathcal{X})$ as follows: T is a maximal torus in G; $B_{\mathbb{C}}$ is a Borel subgroup of $G^\circ_{\mathbb{C}}$ containing $T_{\mathbb{C}}$, with a resulting set of simple roots $\Delta(B_{\mathbb{C}}, T_{\mathbb{C}})$; and $\mathcal{X}$ is a set consisting of a real ray in each complex simple root space (i.e., the set of positive real multiples of some fixed non-0 vector). (Sorry about the pair of modifiers "complex simple".) I will call these 'pinnings', although it doesn't agree with the usual terminology (where we pick individual root vectors, not rays). I claim that $G^\circ/\mathrm{Z}(G^\circ)$ acts simply transitively on the set of pinnings.

Once we have transitivity, freeness is clear: if $g \in G^\circ$ stabilises some pair $(T, B_{\mathbb{C}})$, then it lies in T, and so stabilises every complex root space; but then, for it to stabilise some choice of rays $\mathcal{X}$, it has to have the property that $\alpha(g)$ is positive and real for each simple root α; but also $\alpha(g)$ is a norm-1 complex number, hence trivial, for each simple root α, hence for each root α, so that g is central.

For transitivity, since (1) all maximal tori in G are G°-conjugate, so (2) for every maximal torus T in G, the Weyl group $\mathrm{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ acts transitively on the Borel subgroups of $G^\circ_{\mathbb{C}}$ containing $T_{\mathbb{C}}$, and (3) $\mathrm{W}(G^\circ, T) \to \mathrm{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ is an isomorphism, it suffices to show that all possible sets $\mathcal{X}$ are conjugate. Here's the argument that I came up with to show that they are even T-conjugate; I think it can probably be made much less awkward. Fix a simple root α, and two non-0 elements X_α and X'_α of the corresponding root space. Then there are a positive real number r and a norm-1 complex number z such that $X'_\alpha = rzX_\alpha$. Choose a norm-1 complex number w such that $w^2 = z$. There is then a unique element s_{ad} of $T_{\mathbb{C}}/\mathrm{Z}(G^\circ_{\mathbb{C}})$ such that $\alpha(s_{\mathrm{ad}}) = w$, and $\beta(s_{\mathrm{ad}}) = 1$ for all simple roots $\beta \neq \alpha$. By (4), we can choose a lift s_{sc} of s_{ad} to $(G_{\mathrm{sc}})_{\mathbb{C}} = (G_{\mathbb{C}})_{\mathrm{sc}}$, which necessarily lies in the preimage $(T_{\mathbb{C}})_{\mathrm{sc}}$ of (the intersection with the derived subgroup of) T, and put $t_{\mathrm{sc}} = s_{\mathrm{sc}} \cdot \overline{s_{\mathrm{sc}}}$. Then

$$\alpha(t_{\mathrm{sc}}) = \alpha(s_{\mathrm{sc}})\overline{\overline{\alpha}(s_{\mathrm{sc}})} = \alpha(s_{\mathrm{sc}})\overline{\alpha(s_{\mathrm{sc}})^{-1}} = w \cdot \overline{w^{-1}} = z,$$

and, similarly, $\beta(t_{\mathrm{sc}}) = 1$ for all simple roots $\beta \neq \alpha$. Now the image t of t_{sc} in $G^\circ_{\mathbb{C}}$ lies in $T_{\mathbb{C}}$ and is fixed by conjugation, hence lies in T; and $\mathrm{Ad}(t)X_\alpha = zX_\alpha$ lies on the ray through X'_α.

Since G also acts on the set of pinnings, we have a well defined map $p : G \to G^\circ/\mathrm{Z}(G^\circ)$ that restricts to the natural projection on G°. Now $\ker(p)$ meets every component, but it contains $\mathrm{Z}(G^\circ)$, so it need not be finite. Applying (5) to the Lie group $\ker(p)$ yields the desired subgroup H. Note that, as requested in your improved classification, conjugation by any element of H fixes a pinning, hence, if inner, must be trivial.

□

Remark 1.3. The pinnings show that split reductive algebraic groups look like butterflies!

According to Milne, Grothendieck used the following analogy to talk about the structure theory of algebraic groups. An algebraic group is like a butterfly. The body of the butterfly is the maximal torus. The wings are two opposite Borel subgroups. And the pins are the pinning maps.

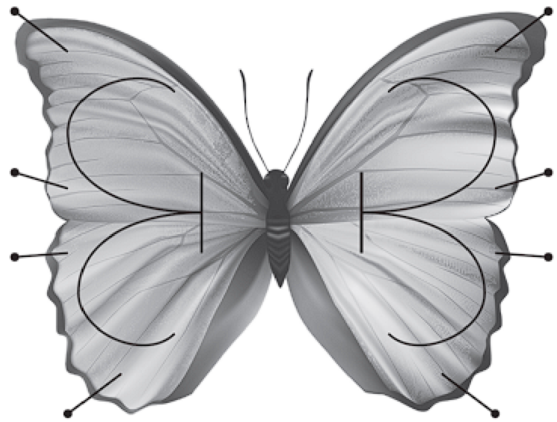


FIGURE 1. Pinning of butterfly

1.3. **Coherent continuation representations.** Until the end of today, G denotes a real reductive group in Harish-Chandra class.

Let's recall some basic definitions of coherent continuation representations, in order for me to do calculations in the case of real classical groups.

Fix a connected reductive complex Lie group $G_{\mathbb{C}}$, together with a Lie group homomorphism $\iota : G \to G_{\mathbb{C}}$ such that its differential $\mathrm{d}\iota : \mathrm{Lie}(G) \to \mathrm{Lie}(G_{\mathbb{C}})$ has the following two properties:

(1) the kernel of $\mathrm{d}\iota$ is contained in the center of $\mathrm{Lie}(G)$;
(2) the image of $\mathrm{d}\iota$ is a real form of $\mathrm{Lie}(G)$.

The analytical weight lattice of $\mathrm{G}_{\mathbb{C}}$ is identified with a subgroup of ${}^a\mathfrak{h}^*$ via $\mathrm{d}\iota$. We write $Q_\iota \subset {}^a\mathfrak{h}^*$ for this subgroup. Denote root lattice by $Q_{\mathfrak{g}}$, weight lattice by $Q^{\mathfrak{g}}$, then

$$Q_{\mathfrak{g}} \subset Q_\iota \subset Q^{\mathfrak{g}}.$$

They are all W stable subgroups. And we have some categories:

(1) $\mathrm{Rep}(\mathfrak{g}, Q_\iota)$ the category of all finite-dimensional representations of $\mathfrak{g}$ whose weight are contained in $Q_{\mathfrak{g}}$, it can be identified with $\mathrm{Rep}(Lie(G_{\mathbb{C}}), Q_\iota)$ and hence $\mathrm{Rep}_{finite}(G_{\mathbb{C}})$;
(2) $\mathrm{Rep}(G)$ the category of all Casselman-Wallach representations of G, and its full subcategories $\mathrm{Rep}_\mu(G)$, $\mathrm{Rep}_S(G)$.

To denote their Grothendieck groups, we replace Rep by $\mathcal{K}$.

Fix a Q_ι-coset $\Lambda = \lambda + Q_\iota \subset {}^a\mathfrak{h}^*$.

Definition 1.4 (coherent family)**.** A $\mathcal{K}(G)$-valued Λ coherent family is a map

$$\Phi : \Lambda \to \mathcal{K}(G)$$

satisfies the following two conditions:

(1) for all $\nu \in \Lambda$, $\Phi(\nu) \in \mathcal{K}_\nu(G)$;

(2) for all representations $F \in \mathrm{Rep}(\mathfrak{g}, Q_\iota)$ and all $\nu \in \Lambda$,

$$F \cdot (\Phi(\nu)) = \sum_{\mu} \Phi(\nu + \mu),$$

where μ runs over all weights of F, counted with multiplicities.

The set of coherent families is denoted by $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$

We have the integral Weyl group $W_\Lambda = \{w \in W \mid w\lambda - \lambda \in Q_\iota\}$ there is a representation of W_Λ on $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$ by

$$(w \cdot \Phi)(\nu) = \Phi(w^{-1}\nu).$$

This is called a coherent continuation representation.

2. Regular Parameters and Outer Automorphisms

2023-12-23

2.1. **Review of regular parameter.** Following yesterday's discussion of coherent continuation representations, we define two basis for it, use Langlands classification, follow [BMSZ22].

Suppose H is a Cartan subgroup of G, by which we mean a subgroup that is the centralizer of a Cartan subalgebra of $\mathrm{Lie}(G)$ (In the case of linear groups, it's just a real algebraic torus.). H has a unique maximal compact subgroup (since G is in Harish-Chandra's class, adjoint representation of H is trivial, H_0 lies in the center of H, as we know, all maximal subgroups of Lie groups are conjugate to each other, use Cartan decomposition of H, we win. In fact, H has a unique maximal compact torus and a unique maximal split torus) We denote by $\Delta_{\mathfrak{h}} \subset \mathfrak{h}^*$ the root system of $\mathfrak{g}$. A root is called imaginary if $\check{\alpha} \in \mathfrak{t}$ (i.e. roots which take pure imaginary values on $\mathrm{Lie}(H)$), an imaginary root α is called compact if $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ is contained in the complexified Lie algebra of a common compact subgroup of G.(this is equivalent to $\mathfrak{g}_\alpha$ belong to a Lie subalgebra of a compact subgroup, since $\mathfrak{g}_\alpha$ lies in a Lie subalgebra of maximal compact subgroup is equivalent to $\mathfrak{g}_{-\alpha}$ Lies in the same subgroup)

There are two important facts about the representation theory of H:

(1) Every Casselman-Wallach representation of H is finite-dimensional, it follows directly from that H is an extension of an abelian group and a finite group;
(2) For every $\Gamma \in \mathrm{Irr}(H)$ differential of Γ is a direct sum of one-dimensional representations attached to a unique $\mathrm{d}\Gamma \in \mathfrak{h}^*$. (it is true because H is in Harish-Chandra's class).

For every Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ containing $\mathfrak{h}$, write

$$\xi_{\mathfrak{b}} : \mathfrak{h} \to {}^a\mathfrak{h},$$

for the linear isomorphism attached to $\mathfrak{b}$ defined by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \to \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}] = {}^a\mathfrak{h}.$$

The transpose inverse of this map is still denoted by $\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \to \mathfrak{h}^*$.

Write

$$W({}^a\mathfrak{h}^*, \mathfrak{h}^*) := \{\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \to \mathfrak{h}^* \mid \mathfrak{b} \text{ is a Borel subalgebra containing } \mathfrak{h}\}.$$

Definition 2.1. For every element $\xi \in W\left({}^{a}\mathfrak{h}^{*}, \mathfrak{h}^{*}\right)$, put

$$\delta(\xi) := \frac{1}{2} \cdot \sum_{\alpha \text{ is an imaginary root in } \xi\Delta^{+}} \alpha - \sum_{\beta \text{ is an compact imaginary root in } \xi\Delta^{+}} \beta \in \mathfrak{h}^{*}.$$

Write $\mathcal{P}_{\Lambda}(G)$ for the set of all triples $\gamma = (H, \xi, \Gamma)$, where H is a Cartan subgroup of G, $\xi \in W\left({}^{a}\mathfrak{h}^{*}, \mathfrak{h}^{*}\right)$, and

$$\begin{array}{rccc} \Gamma\colon & \Lambda & \longrightarrow & \mathrm{Irr}(H) \\ & \nu & \longmapsto & \Gamma_{\nu} \end{array}$$

is a map with the following properties:

(1) $\Gamma_{\nu+\beta} = \Gamma_{\nu} \otimes \xi(\beta)$ for all $\beta \in Q_{\iota}$ and $\nu \in \Lambda$;
(2) $\mathrm{d}\Gamma_{\nu} = \xi(\nu) + \delta(\xi)$ for all $\nu \in \Lambda$.(note that Cartan subgroups not always have irr repn with this differential!)

Here $\xi(\beta)$ is naturally viewed as a character of H by using the homomorphism $\iota : H \to H_{\mathbb{C}}$, and $H_{\mathbb{C}}$ is a Cartan subgroup of $G_{\mathbb{C}}$ containing $\iota(H)$.

2.2. **Outer automorphism of algebraic groups.**

Lemma 2.2 (a simple discovery of abstract algebra)**.** *G an abstract group, H a normal subgroup of G, if G acts transitively on a set X such that the restriction of this action to H is free and transitive, then the short exact sequence of groups*

$$1 \to H \to G \to G/H \to 1$$

Proof. Actually choose an element $x \in X$, we have

$$G \simeq \mathrm{Stab}_{G}(x) \ltimes H.$$

□

Using this lemma, we can see that for a complex connected reductive algebraic group G, $\mathrm{Aut}(G)$ acts on the set of pinnings transitively, with its normal subgroup the $\mathrm{Inn}(G)$ acts freely and transitively, so every pinning gives us a split of the short exact sequence

$$1 \to \mathrm{Inn}(G) \to \mathrm{Aut}(G) \to \mathrm{Out}(G) \to 1.$$

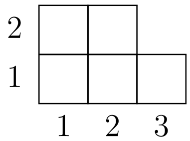
Some facts about split connected reductive algebraic groups over arbitrary field k:

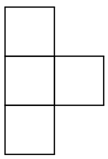
(1) $\{\text{split connected reductive algebraic groups}\}/_{\backsim} = \{\text{root data}\}/_{\backsim}$;
(2) For any two automorphisms of G, if they induce the same automorphism of root datum, then they differ by an inner automorphism;
(3) Fix a pinning, we get an isomorphism $\mathrm{Out}(G) \simeq \mathrm{Aut}(D_b)$.

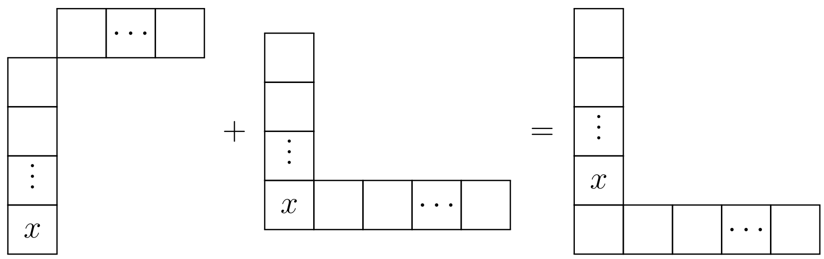
3. Young Diagrams and Discrete Series

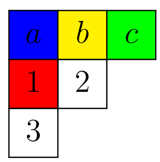
2024-02-27

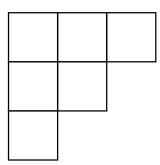
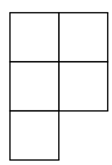
Some code examples of Young diagrams and tableaux using the `ytableau` package:

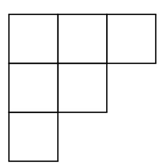


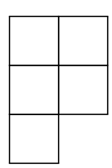


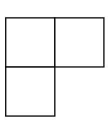


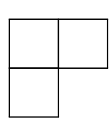


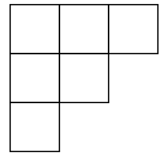
 

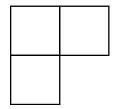




ce est chinois  english

$a =$  $+ b$

youngdiagram1: 

young2: 

For a compact connected Lie group G, choose a maximal torus T, there is a canonical 1-1 correspondence:

$$X^*(T)/W \longleftrightarrow (X^*(T)+\rho)/W \longleftrightarrow \mathrm{Irr}(G)$$

Moreover, if we choose a positive system Ψ^+, and then ρ_{Ψ^+}, the above sets also canonically isomorphic to the set of dominant analytically integral weights:

$$\{\lambda \in X^*(T) \,|\, \langle \lambda, \alpha\rangle \geq 0, \text{for all } \alpha \in \Psi^+\}$$

This can be generalized to parameter of discrete series representations for connected semisimple Lie groups G with compact Cartan subgroup T :

$$(X^*(T)+\rho)^{reg}/W_c$$

If a discrete series representation V has minimal K type which has maximal weight ν, then it's parameter is $\lambda = \nu - \rho + 2\rho_c$.(not sure)

Note that for a fixed infinitesimal character, there are exactly $|W/W_c|$ different discrete series representations.

4. Continuous complex characters

2024-02-28

Some facts about continuous complex characters of some common abelian topological groups:

field	character group	correspondence
$(\mathbb{R}_+^\times, \times)$	$\mathbb{C}$	$\xi \mapsto (t \mapsto t^\xi)$
$(\mathbb{C}^\times, \times)$	$\mathbb{C}\times\mathbb{Z}$	$(\xi, n) \mapsto (z \mapsto \lvert z\rvert^\xi (\frac{z}{\lvert z\rvert})^n)$
$(\mathbb{S}^1, \times)$	$\mathbb{Z}$	$n \mapsto (z \mapsto z^n)$
$(\mathbb{Q}_p, +)$	$\mathbb{Q}_p$	$\xi \mapsto (x \mapsto \exp(2\pi i \xi \{x\}_p))$

Remark 4.1. Note that all continuous characters of $\mathbb{Q}_p$ are automatically locally constant (smooth), which is consistent with the case of real lie groups.

5. Associated Varieties of Casselman-Wallach Representations

2024-10-22

Lemma 5.1. *The associated variety of an irreducible Casselman-Wallach representation of a real reductive group in Harish-Chandra class is the closure of a nilpotent orbit in $\mathfrak{g}^*$.*

Proof. By the work of Borho, Brylinski [BB82], and Joseph [Jos85], the associated variety of any primitive ideal (annihilator of an irreducible $\mathfrak{g}$-module) is the closure of a single nilpotent orbit. So we only need to prove that $\mathrm{gr}(\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V))$ is a primitive ideal.

Since the K-finite part V_K is dense in V, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V) = \mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V_K)$, we only need to consider the annihilator of the irreducible $(\mathfrak{g}, K)$-module V_K.

If G is connected, then K is also connected, so V_K is irreducible as $\mathcal{U}(\mathfrak{g})$-module, in this case, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V)$ is a primitive ideal.

If G is not connected, let G_0 be the identity component of G, $K_0 = G_0 \cap K$ is a maximal compact subgroup of G_0, then $V_K|_{G_0}$ is a finite length $(\mathfrak{g}, K_0)$-module, it has an irreducible quotient $V_K \to V_0$, then according to the theory of induction of $(\mathfrak{g}, K)$-modules [KV95, Chapter 2] we have an injective morphism of $(\mathfrak{g}, K_0)$-modules:

$$V_K|_{K_0} \hookrightarrow (\mathrm{induced}_{K_0}^{K}(V_0))|_{K_0} \cong \bigoplus_{\text{double cosets } K_0 k K_0} kV_0, \tag{5.1}$$

where kV_0 be the $(\mathfrak{g}, K_0)$-module whose underlying space is V_0, whose action by $X \in \mathfrak{g}$ is the usual action by $\mathrm{Ad}(k)X$, whose action by $g \in K_0$ is the usual action by $k^{-1}gk$, it has the same annihilator in $\mathcal{U}(\mathfrak{g})$ as V_0 since G is in Harish-Chandra's class, and hence the annihilator of V_K is the same as the annihilator of V_0, which is a primitive ideal.

□

6. Split reductive algebraic groups over arbitrary field

2024-11-28

Any split connected reductive group over arbitrary field can be uniquely defined and split over $\mathbb{Z}$. (Chevalley group)

Classification of split reductive group is the same over arbitrary non-empty schemes, which is equivalent to the classification of root data.(cf. [SGA3])

7. Geometric Pre-quantization

2024-12-17

Geometric Pre-quantization

Let X be a smooth manifold. There are natural maps

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{\mathrm{dR}}(X; \mathbb{R}) \simeq \mathrm{H}^2_{\check{\mathrm{C}}\mathrm{ech}}(X; \mathbb{R}).$$

Construction of the first map: For a (complex) line bundle $\mathbb{L} \in \mathrm{Pic}^h(X)$, let $\langle , \rangle$ be a Hermitian metric on it and ∇ be a unitary connection. (Every line bundle $\mathbb{L} \in \mathrm{Pic}(X)$ has a Hermitian metric and a unitary connection on it.)

- choose a locally trivialization $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- choose Unitary local frames $\{s_\alpha\}$ for $\mathbb{L}$ on each U_α, then $s \leftrightarrow (f_\alpha)$ and $\nabla \leftrightarrow (\theta_\alpha)$ (unitary implies that θ_α are pure imaginary);
- define $\Omega := \mathrm{d}\theta_\alpha$, and $c_1(\mathbb{L}) := [\frac{1}{2\pi i}\Omega] \in \mathrm{H}^2_{dR}(X; \mathbb{R})$ ($\Omega = \mathrm{curv}\nabla$).

$c_1(\mathbb{L})$ is independent of the choice of $\langle , \rangle$ and ∇ by Chern-Weil theory. And we can see that $\theta_\alpha - \theta_\beta = \frac{\mathrm{d}g_{\alpha,\beta}}{g_{\alpha,\beta}} = \mathrm{d}\log(g_{\alpha,\beta})$ on $U_\alpha \cap U_\beta$, where $g_{\alpha,\beta}$ is the translation function ($s_\beta = g_{\alpha,\beta} s_\alpha$).

Construction of the second map (classical de-Rham isomorphism), choose a 2-form $\omega \in \mathrm{H}^2_{dR}(X; \mathbb{R})$

- choose a cover $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- on each U_α, choose a 1-form θ_α such that $\mathrm{d}\theta_\alpha = \omega|_{U_\alpha}$;
- since $\mathrm{d}\theta_\alpha = \mathrm{d}\theta_\beta$ on $U_\alpha \cap U_\beta$, we can choose smooth function $c_{\alpha,\beta}$ such that $\mathrm{d}c_{\alpha,\beta} = \theta_\alpha - \theta_\beta$;

- let $c_{\alpha,\beta,\gamma} = c_{\alpha,\beta} + c_{\beta,\gamma} + c_{\gamma,\alpha}$, which is a constant, and we get the corresponding 2-cocycle $(c_{\alpha,\beta,\gamma}) \in \mathrm{H}^2_{\text{Čech}}(X;\mathbb{R})$.

If ω come from the first map, we can choose $c_{\alpha,\beta} = \frac{1}{2\pi i}\log g_{\alpha,\beta}$, and hence $c_{\alpha,\beta,\gamma} = \frac{1}{2\pi i}(\log g_{\alpha,\beta} + \log g_{\beta,\gamma} + \log g_{\gamma,\alpha}) \in \mathbb{Z}$. So the morphism is in fact

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{dR}(X;\mathbb{Z}) \simeq \mathrm{H}^2_{\text{Čech}}(X;\mathbb{Z}).$$

In fact the first map is also an isomorphism. (cf. prequantization) For a 2-form ω, choose θ_α and $c_{\alpha,\beta}$ as above, let $g_{\alpha,\beta} = \exp(2\pi i c_{\alpha,\beta})$, these are transition functions of a complex line bundle $\mathbb{L}$.

Corollary 7.1. *Assume that G is a connected Lie group. The following are equivalent:*

- *$\Omega \subseteq \mathfrak{g}^*$ is integral ($[\sigma_\Omega] \in \mathrm{H}^2(X;\mathbb{Z})$);*
- *there exists a G-equivariant complex line bundle over Ω with a G-invariant Hermitian connection ∇ such that*
$$\frac{1}{2\pi i}\mathrm{curv}\,(\nabla) = \sigma_\Omega.$$

Theorem 2.6 (Weil)**.** *Let M be a smooth manifold and ω a real, closed 2-from whose cohomology class $[c]$ is integral. Then there is a unique Hermitian line bundle $\mathbb{L}$ over M with unitary connection ∇ so that $c_1(\mathbb{L}) = [c]$.*

Sketch of Proof. Existence: Reverse the arguments above. Since $[c]$ is integral, $e^{2\pi c_{ijk}} = 1$. As a consequence, $g_{ij}g_{jk}g_{kl} = 1$. So g_{ij}'s are the transition function for some line bundle whose first Chern class is $[c]$.

Uniqueness: Suppose $c_1(\mathbb{L}) = c_1(\widetilde{\mathbb{L}})$. Let $h_{ij} = \frac{1}{2\pi i}g_{ij}$ and define $\tilde{h}_{ij}$ similarly. Then the functions $\hat{h}_{ij} = h_{ij} - \tilde{h}_{ij}$ satisfies the relation

$$\hat{h}_{ij} + \hat{h}_{jk} + \hat{h}_{ki} = 0.$$

We take a partition of unity ρ_k and let $\lambda_i = e^{2\pi i \sum \hat{h}_{ki}\rho_k}$. Then

$$\lambda_i g_{ij} \lambda_j^{-1} = e^{2\pi i (h_{ij} + \sum(\hat{h}_{ki} - \hat{h}_{kj})\rho_k)} = e^{2\pi i (h_{ij} - \hat{h}_{ij})} = \tilde{g}_{ij}.$$

So as line bundles $\mathbb{L} \simeq \widetilde{\mathbb{L}}$. □

FIGURE 2. Proof

8. Tianyuan Problems [2]

2025-08-14

Let G be a finite central extension of an open subgroup of $\mathbb{R}$-points of a connected reductive algebraic group (reductive Nash group [Sun15]) with Lie algebra $\mathfrak{g}$. Fix an invariant nondegenerate bilinear form $B(-,-)$ on $\mathfrak{g}$.

For an irreducible Casselman-Wallach representation π of G, let $\mathrm{Wh}_{\mathcal{O}}(\pi)$ be the space of generalized Whittaker models of π associated to the nilpotent orbit $\mathcal{O}$. Let

$$\mathrm{WO}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a nilpotent orbit such that } \mathrm{Wh}_{\mathcal{O}}(\pi) \neq 0\},$$

and the *Whittaker support* $\mathrm{WS}(\pi)$ of π be the set of all maximal orbits in $\mathrm{WO}(\pi)$ with respect to closure inclusion:

$$\mathrm{WS}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a maximal orbit in } \mathrm{WO}(\pi)\}.$$

Another nilpotent invariant is the wavefront cycle:

$$\mathcal{WF}(\pi) = \sum_i m_i[\mathcal{O}_i].$$

With the corresponding wavefront set :

$$\mathrm{WF}(\pi) = \bigcup_{m_i \neq 0} \overline{\mathcal{O}_i}.$$

Mœglin-Waldspurger [MW87] proved that $\mathrm{WF}(\pi) = \mathrm{WS}(\pi)$ when F is a p-adic field. We want to test this conjecture for arbitrary reductive almost linear Nash groups.

Conjecture 8.1 (Main conjecture)**.** *For any reductive almost linear Nash group G and any irreducible Casselman-Wallach representation π:*

$$\mathrm{WF}(\pi) = \mathrm{WS}(\pi).$$

Take a nilpotent orbit $\mathcal{O} \subseteq \mathfrak{g}$ (or $\mathfrak{sl}_2$-triple $\gamma = \{Y, H, X\}$ in $\mathfrak{g}$). $\mathfrak{g}/\mathfrak{g}^Y$ is a symplectic space with symplectic form $\langle -,-\rangle := B(Y,[-,-])$. Set $G^Y := \mathrm{Stab}_G(Y)$, there is a natural homomorphism $G^Y \to \mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,-\rangle)$. Let $\widetilde{G^Y}$ be the double cover of G^Y induced by the metaplectic double cover of $\mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,-\rangle)$.

Definition 8.2. The nilpotent $\mathcal{O}$ is called admissible if the double cover $\widetilde{G^Y}$ is split over the identity component of $(G^Y)_0$. It is called quasi-admissible if $\widetilde{G^Y}$ has a finite-dimensional genuine representation.

Result of [GGS21] says that $\mathrm{WS}(\pi)$ consists of quasi-admissible orbits. But they do not know whether the non-admissible quasi-admissible orbits appear in Whittaker supports of representations. We may ask the same question for the wavefront set:

Problem 1: Are all the nilpotent orbits in the wavefront cycle admissible?

For the case when G is general linear, it is well known that the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G consists of a single nilpotent adjoint orbit. Together with some counting results, it maybe possible to construct all irreducible representations of general linear group via theta correspondence and coherent continuation.

[2] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

Problem 2: Can we construct all irreducible representations via theta correspondence, coherent continuation, and twist of characters?

Assuming the construction steps above are done, it then remains to study the Whittaker support $\mathrm{WS}(\pi)$ of π. We focus on two types of interactions with Whittaker models - theta correspondence and coherent continuation.

On one hand, Gomez-Zhu [GZ14] proved for type I reductive (symplectic-orthogonal) dual pairs in the stable range that the big theta lift preserves the structure of Whittaker models:

$$\mathrm{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \mathrm{Wh}_{\mathcal{O}}(\pi^\vee).$$

Noting that $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a (reductive) dual pair for any $n, m \in \mathbb{N}$, one observes that if one can prove an analogue of Gomez-Zhu's result for type II dual pairs, one can use it to infer properties of the structure of the Whittaker support $\mathrm{WS}(\pi)$. Hence we propose the following subconjecture:

Conjecture 8.3. *If $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a type II reductive dual pair over F in the stable range (with $n \leq m$), and π is a genuine Casselman-Wallach representation of G, then*

$$\mathrm{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \mathrm{Wh}_{\mathcal{O}}(\pi^\vee),$$

where $\pi^\vee$ denotes the smooth contragredient of π. Furthermore, the above will imply that

$$\mathrm{WS}(\Theta(\pi)) = \mathrm{WS}(\pi^\vee).$$

On the other hand, Barbash-Vogan [BV80] proved that under coherent continuation, the multiplicity of wavefront cycle changes in the form of a Goldie rank polynomial, i.e.

$$\mathcal{WF}(\pi(\lambda)) = p(\lambda) \cdot (\sum_i c_i[\mathcal{O}_i]),$$

where $p(\lambda)$ is the goldie rank polynomial, λ is dominant. We expect some similar result for the multiplicity of the generalized Whittaker model.

Problem 3: For a fixed nilpotent orbit $\mathcal{O}$, does there exist a constant $c_{\mathcal{O}}$ such that $\dim \mathrm{Wh}_{\mathcal{O}}(\pi(\lambda)) = c_{\mathcal{O}} \cdot p(\lambda)$?

9. Orbit Method and Wavefront Sets

2025-10-01

Orbit method for nilpotent Lie groups.

Let G be a connected, simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$.

We have Fourier transform:

$$\mathcal{F} : \mathcal{S}(\mathfrak{g})\mathrm{d}X \to \mathcal{S}(\sqrt{-1}\mathfrak{g}^*), \quad f\mathrm{d}X \mapsto \widehat{f\mathrm{d}X}, \quad \widehat{f\mathrm{d}X}(l) = \int_{\mathfrak{g}} f(X)e^{\langle l, X\rangle}\mathrm{d}X.$$

Take its dual:

$$\mathcal{S}'(\sqrt{-1}\mathfrak{g}^*) \to (\mathcal{S}(\mathfrak{g})\mathrm{d}X)', \quad T \mapsto \check{T}, \quad \langle \check{T}, f\mathrm{d}X\rangle = \langle T, \widehat{f\mathrm{d}X}\rangle.$$

For any irreducible unitary representation π of G, we have its character $\Theta_\pi \in C^{-\infty}(G)$ defined by

$$\Theta_\pi(f) = \mathrm{Tr}\left(\pi\left(f\mathrm{d}g\right)\right), \quad f\mathrm{d}g \in C_c^\infty(G)\mathrm{d}g.$$

Using the exponential map $\exp : \mathfrak{g} \to G$ (isomorphism in this case), we can view Θ_π as a distribution on $\mathfrak{g}$:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(f \circ \exp), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g})\mathrm{dX}.$$

In fact, θ_π is a tempered generalized function on $\mathfrak{g}$ (it can continuously extends to $\mathcal{S}(\mathfrak{g})\mathrm{dX}$). By the work of Howe, Kirillov, and others, we have the following theorem:

Theorem 9.1. *There exists a unique coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$ such that $\check{\theta}_\pi = \mu_{\mathcal{O}}$, where $\mu_{\mathcal{O}}$ is the canonical measure on $\mathcal{O}$.*

This defines a bijection between $\widehat{G}$ (the unitary dual of G) and the set of coadjoint orbits in $\sqrt{-1}\mathfrak{g}^*$.

Conversely, for any coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$, we can construct an irreducible unitary representation $\pi_{\mathcal{O}}$ of G such that $\check{\theta}_{\pi_{\mathcal{O}}} = \mu_{\mathcal{O}}$. This is the more interesting part, use the method of geometric quantization.

Take any element $\lambda \in \mathcal{O}$, we have an identification $\mathcal{O} = G/G_\lambda$, $lambda : \mathfrak{g}_\lambda \to \mathbb{C}$ is a Lie algebra homomorphism, and hence $\chi_\lambda : G_\lambda \to \mathbb{C}^\times$ defined by $\chi_\lambda(\exp X) = e^{\langle \lambda, X\rangle}$ is a unitary character of G_λ. Then we can defined a Hermitian line bundle $\mathcal{L}_\lambda = G \times^{G_\lambda} \mathbb{C}$ over $\mathcal{O}$, with a natural G-invariant connection ∇ such that $\mathrm{curv}(\nabla) = \omega_{\mathcal{O}}$, where $\omega_{\mathcal{O}}$ is the Kirillov-Kostant-Souriau symplectic form on $\mathcal{O}$.

Then we can define the Hilbert space of square integrable sections of $\mathcal{L}_\lambda$, but this space is too large. We need to choose a polarization $\mathfrak{h} \subseteq \mathfrak{g}_{\mathbb{C}}$ (maximal isotropic subalgebra with respect to the bilinear form $(X, Y) \mapsto \langle \lambda, [X, Y]\rangle$) and consider the space of square integrable holomorphic sections of $\mathcal{L}_\lambda$ which are covariant constant along $\mathfrak{h}$.

In fact the polarization $\mathfrak{h}$ corresponds to a G-invariant integrable distribution on $\mathcal{O}$, its a subbundle P of $T\mathcal{O}$ each fiber is a lagrangian subspace. Then we can define the space of polarized sections:

$$\Gamma_P(\mathcal{O}, \mathcal{L}_\lambda) = \{s \in \Gamma^\infty(\mathcal{O}, \mathcal{L}_\lambda \otimes |\omega_{\mathcal{O}/\mathcal{P}}|^{\frac{1}{2}}) \mid \nabla_X s = 0, \forall X \in P\}.$$

On this space, we have a unitary G action and get a unitary representation $\pi_{\mathcal{O}}$ of G, it turns out that $\pi_{\mathcal{O}}$ is irreducible and independent of the choice of the polarization $\mathfrak{h}$.

Kirillov's philosophy works well for nilpotent Lie groups, but for more general Lie groups, it is not true anymore.

Now assume G is a real reductive group, with Lie algebra $\mathfrak{g}_0$. For any irreducible Casselman-Wallach representation π of G, we still have its generalized function character $\Theta_\pi \in C^{-\infty}(G)$, and hence a tempered generalized function θ_π on $\mathfrak{g}$. Here $\theta_\pi \in C^{-\infty}(\mathfrak{g}_0)$ is defined as follows:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(\exp_*(\sqrt{\det(\exp)}f\mathrm{dX})), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g}_0)\mathrm{dX}.$$

The generalized function Θ_π has the following properties:

- Θ_π is invariant under conjugation of G;
- Θ_π is an eigendistribution of $\mathcal{U}(\mathfrak{g})^G$;
- Θ_π is locally integrable, and real analytic on the regular semisimple locus G^{rs}.

By the work of Rossmann, we have the following theorem:

Theorem 9.2. *If π is tempered, then*

$$\theta_\pi|_{nbhd\ of\ 0} = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}|_{nbhd\ of\ 0}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$*, is the orbital integral along* $\mathcal{O}_i$*.*

For an arbitrary generalized function $\Theta \in C^{-\infty}(M)$, we can define its wavefront set $\mathrm{WF}(\Theta) \subseteq \sqrt{-1}T^*M$ as follows:

Definition 9.3. For any $(x, \xi) \in \sqrt{-1}T^*M$, $(x, \xi) \notin \mathrm{WF}(\Theta)$ if there exists a compactly supported smooth function φ on M such that $\varphi(x) \neq 0$, and an open conic neighborhood Γ of ξ such that for any $N \in \mathbb{N}$, there exists a constant $C_N > 0$ such that

$$|\widehat{\varphi\Theta}(t\eta)| \leq C_N t^{-N}, \quad \forall t > 0, \eta \in \Gamma.$$

$\mathrm{WF}(\Theta)$ is a closed conic subset of $\sqrt{-1}T^*M$ (singular direction of Θ).

In particular, we can define the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G as $\mathrm{WF}(\Theta_\pi) \cap (\sqrt{-1}T_1^*(G))$, it is a closed conic G-invariant subset of $\sqrt{-1}\mathfrak{g}_0^*$. The GK-dimension of π is defined as

$$\mathrm{Dim}(\pi) = \frac{1}{2}\dim(\mathrm{WF}(\pi)).$$

θ_π is a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, and

$$\theta_\pi^*(X) := \lim_{t \to 0} t^{\mathrm{Dim}(\pi)}\theta_\pi(tX)$$

is still a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, where $\mathcal{B}$ is the flag variety of G. We can consider it as in $C^{-\infty}(\mathfrak{g}_0)$.

Theorem 9.4. *(Barbasch-Vogan) The asymptotic of* θ_π *defined above is a non-negative linear combination of Fourier transforms of nilpotent orbital integrals:*

$$\theta_\pi^* = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$*, is the orbital integral along* $\mathcal{O}_i$*.*

$\mathrm{WF}(\pi) = \bigcup_{c_i \neq 0} \overline{\mathcal{O}_i}$.

10. Conormal Bundles and Characteristic Cycles

2025-10-02

10.1. **Conormal bundle of singular subvariety.** Let $\mathcal{B}$ be the flag variety of G. $\overline{\mathcal{B}_y} \subsetneq \overline{\mathcal{B}_w}$ two schubert varieties, what is the relation between their conormal bundles $T^*_{\mathcal{B}_y}\mathcal{B}$ and $T^*_{\mathcal{B}_w}\mathcal{B}$?

Let X be a smooth algebraic variety over $\mathbb{C}$, $Z \subseteq X$ be a closed subvariety, define the conormal bundle T_Z^*X of Z in X as follows:

$$T_Z^*X = \overline{T^*_{Z^{\mathrm{reg}}}X}.$$

For a singular point $z \in Z$ what is the fiber of T_Z^*X at z?

Here is a simple example:

Example 10.1. Let $X = \mathbb{C}^2$ and

$$Z = \{xy = 0\} \subset X$$

be the union of the coordinate axes.

- The regular locus is

$$Z^{\mathrm{reg}} = (x\text{-axis} \setminus \{0\}) \cup (y\text{-axis} \setminus \{0\}).$$

- The singular locus is
$$S := Z \setminus Z^{\mathrm{reg}} = \{0\}.$$

Now, the fibre of the conormal bundle at the origin is
$$(T^*_Z X)_0 = \mathrm{Ann}(T_{x\text{-axis}}) \cup \mathrm{Ann}(T_{y\text{-axis}}) = \langle dy \rangle \cup \langle dx \rangle,$$
which is a union of two lines in $T^*_0\mathbb{C}^2$.

On the other hand,
$$(T^*_S X)_0 = (T^*_{\{0\}} X)_0 = T^*_0 X,$$
the entire cotangent plane.

Thus,
$$(T^*_Z X)_0 \subsetneq (T^*_{Z-Z^{\mathrm{reg}}} X)_0,$$

10.2. **Summary of paper [KT84, The characteristic cycles of holonomic systems on a flag manifold].** Let G be a semisimple simply connected complex algebraic group. Define Steinberg variety:

$$Z = \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathrm{Lie}(B') \cap \mathrm{Lie}(B'')\} (= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, \}).$$
$(= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathfrak{n}'^{\perp} \cap \mathfrak{n}''^{\perp}\}$ more naturally.)

Here $\mathcal{N} \subseteq \mathfrak{g}$ is the nilpotent cone of $\mathfrak{g}$, $\mathcal{B}$ is the flag variety of G. It can be identify with the union of conormal bundles of all G-orbits in $\mathcal{B} \times \mathcal{B}$:
$$Z = \bigcup_{w \in W} T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B}),$$
which is a closed conic lagrangian subvariety of $T^*(\mathcal{B} \times \mathcal{B})$.

We also have the moment map:
$$\mu : Z \to \mathcal{N}, \quad (x, B', B'') \mapsto x.$$

By taking characteristic cycles, we get a map from the (complex) Grothendieck group of G-equivariant holonomic D-modules on $\mathcal{B} \times \mathcal{B}$ to the top Borel-Moore homology of Z:
$$\mathrm{CC} : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}).$$
We also have isomorphisms:
$$h : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to \mathbb{C}[W], \quad \mathcal{M}_w \mapsto w, \quad \mathcal{L}_w \mapsto a(w)$$

By Kazhdan-Lusztig, we have:
$$a(w) = \sum_{y \le w} (-1)^{l(y)+l(w)} P_{y,w}(1) y.$$

There is a convolution product on $K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}}))$ which makes h an isomorphism of algebras.

On
$$H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{w \in W} H^{\mathrm{BM}}_{\mathrm{top}}(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}, \mathbb{C}) \simeq \bigoplus_{w \in W} \mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B})}],$$
In the paper of Kazhdan-Lusztig [KL80], they use topological method to construct a $W \times W$ action on $H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C})$.

Moreover there is a $W\times W$ module decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z,\mathbb{C})\simeq\bigoplus_{\mathcal{O}}H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}},\mathbb{C}),$$

where $\mathcal{O}$ runs over all nilpotent orbits in $\mathfrak{g}^*$, $Z_{\mathcal{O}}=\mu^{-1}(\mathcal{O})$. There is also an isomorphism of $W\times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}},\mathbb{C})\simeq(H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x,\mathbb{C})\otimes H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x,\mathbb{C}))_{A_G(x)},$$

($A_G(x)$ coinvariants) where $x\in\mathcal{O}$, $\mathcal{B}_x=\{B'\in\mathcal{B}\,|\,x\in\mathrm{Lie}(B')\}$ (the Springer fiber), $A_G(x)=\mathrm{Stab}_G(x)/\,\mathrm{Stab}_G(x)^0$.

By classical Springer theory, $H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x,\mathbb{C})$ is a representation of $W\times A_G(x)$, and we have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x,\mathbb{C})\simeq\bigoplus_{\rho\in\widehat{A_G(x)}}\tau(x,\rho)\otimes\rho.$$

Here $\tau(x,\rho)$ is an irreducible representation of W or zero.

So we have a decomposition of $W\times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}},\mathbb{C})\simeq\bigoplus_{\rho\in\widehat{A_G(x)}}\tau(x,\rho)\otimes\tau(x,\rho).$$

Combine the above together, we finally have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}},\mathbb{C})\simeq\bigoplus_{\rho\in\widehat{A_G(x)}}\tau(x,\rho)\otimes\tau(x,\rho),\quad H^{\mathrm{BM}}_{\mathrm{top}}(Z,\mathbb{C})\simeq\bigoplus_{\mathcal{O}}H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}},\mathbb{C}).$$

For any $w\in W$, there is a unique nilpotent orbit $\mathrm{St}(w)$ such that $\mu(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})})=\overline{\mathrm{St}(w)}$. The map $w\mapsto\mathrm{St}(w)$ defines a surjective(?) map from W to the set of nilpotent orbits in $\mathfrak{g}^*$ (this map can be computed via Robinson-Schensted algorithm).

Corollary 10.2. *For any nilpotent orbit $\mathcal{O}$, there is an isomorphism of $W\times W$-modules:*

$$\bigoplus_{w\in W,\mathrm{St}(w)=\mathcal{O}}\mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})}]\simeq\bigoplus_{\rho\in\widehat{A_G(\mathcal{O})}}\tau(\mathcal{O},\rho)\otimes\tau(\mathcal{O},\rho).$$

Theorem 10.3. *(Main theorem of [KT84]) The map*

$$\mathrm{CC}:K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}}))\to H^{\mathrm{BM}}_{\mathrm{top}}(Z,\mathbb{C}),$$

is an isomorphism and $W\times W$-equivariant.

Define $b(w)=h\circ\mathrm{CC}^{-1}([\overline{T^*_{\mathcal{Y}_w}\mathcal{B}\times\mathcal{B}}])$. Now we have three basis of $\mathbb{C}[W]$:

$$\{w\}_{w\in W},\quad\{a(w)\}_{w\in W},\quad\{b(w)\}_{w\in W}.$$

It was conjectured by Kashiwara and Tanisaki that: $a(w)=b(w)$ when $G=\mathrm{SL}_n(\mathbb{C})$ which means the characteristic cycle of $\mathcal{L}_w$ is irreducible, but unfortunately it is not true. Counterexample exists when $G=\mathrm{SL}_8(\mathbb{C})$, where Kashiwara-Saito singularities appears in some Schubert varieties.

11. Springer Representations and Characteristic Cycles

2025-10-05

The Springer representation

Let G be a connected complex reductive group, K a symmetric subgroup of G. We can define the Springer representation of W on $H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q})$.

Recall that we have the following diagram:

$$\begin{tikzcd} \widetilde{\mathcal{N}} \arrow[r, hook] \arrow[d, two heads, "p_{\mathcal{N}}"] & \widetilde{\mathfrak{g}} \arrow[d, two heads, "p"] & \widetilde{\mathfrak{g}^{\mathrm{rs}}} \arrow[l, hook'] \arrow[d, two heads, "p_{\mathrm{rs}}"] \\ \mathcal{N} \arrow[r, hook] & \mathfrak{g} & \mathfrak{g}^{\mathrm{rs}} \arrow[l, hook'] \end{tikzcd}$$

In this diagram p_{rs} is a W-torsor. So we have a natural W-action on $p_{\mathrm{rs},*}\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$ and hence on $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L})$, where $\mathcal{L} = (p_{\mathrm{rs}})_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$. There is a canonical isomorphism of perverse sheaves $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L}) \simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[\dim(\mathfrak{g})]$, so there is a natural W-action on $\mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}$. By proper base change, we have

$$\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}} \simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}})|_{\mathcal{N}}. \quad (\mathrm{Spr} = \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[\dim \mathcal{N}]\text{called the Springer sheaf})$$

So there is a natural W-action on $\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}$ (actually $\mathrm{End}_{\mathrm{Perv}(\mathcal{N})}(\mathrm{Spr}) = \mathbb{Q}[W]$) and hence on $H_c^\bullet(T^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathcal{N}, \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$ and $H_c^\bullet(\mathcal{B}_x, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\{x\}, \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$. (action on Springer fiber)

Since $T_K^*\mathcal{B} = p_{\mathcal{N}}^{-1}(\mathfrak{p})$, we also have a natural W-action on

$$H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathfrak{p} \cap \mathcal{N}, (\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})|_{\mathfrak{p}\cap\mathcal{N}}).$$

Finally, there is a W-action on $(H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = \bigoplus_{\mathcal{O} \in K\backslash\mathcal{B}} \mathbb{Q}[\overline{T_{\mathcal{O}}^*\mathcal{B}}]$.

Theorem 11.1 (Tanisaki)**.** *The map:*

$$\mathrm{CC} : K_0(\mathrm{Mod}_K^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B}})) \to (H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = H_{2d}^{\mathrm{BM}}(T_K^*\mathcal{B}, \mathbb{Q}),$$

is surjective and W-equivariant.

12. Holonomic Systems and Characteristic Cycles

2025-10-09

$X = \mathrm{Spec}(R)$ smooth affine with coordinate $x_1, \ldots, x_n$, M a finitely generated $\mathcal{D}_X$-module.

Definition 12.1. M is called holonomic if $\dim(\mathrm{Ch}(M)) = n$.

Corollary 12.2. *If M is holonomic, then* $\mathrm{Ch}(M)$ *is lagrangian in* T^*X*, and* $\mathrm{CC}(M) = \sum_{E_i \subseteq X, irr} n_i[\overline{T_{E_i}^*X}]$.

The de-Rham complex and analytification:

$$\mathcal{K}(M; \partial_1) = [\, M \xrightarrow{\partial_1} M \,],$$

$$\mathcal{K}(M; \partial_1, \partial_2) = [\, M \xrightarrow{(\partial_1, \partial_2)} M \oplus M \xrightarrow{(\partial_2, -\partial_1)} M \,].$$

We can define $\mathcal{K}(M; \partial_1, \ldots, \partial_n)$ inductively.

There is another way to define the de-Rham complex, independent of the choice of coordinates:

$$\mathrm{DR}(M) = [M \xrightarrow{\nabla} \Omega_X^1 \otimes M \xrightarrow{\nabla} \cdots \xrightarrow{\nabla} \Omega_X^n \otimes M].$$

Example 12.3. Let $X = \mathbb{C}$, $M = \mathbb{C}[x, x^{-1}] \cdot x^\alpha$, $\alpha \in \mathbb{C}$, $\partial_x x^\alpha = \alpha \frac{1}{x} x^\alpha$. Then $M_\alpha = \mathcal{D}_X / \mathcal{D}_X (x\partial_x - (\alpha - k))$. To get the *multiplicities* (the characteristic cycle), we examine the primary decomposition of the defining principal-symbol ideal:

$$(x\xi) = (x) \cap (\xi).$$

Each factor appears with exponent 1, and the associated graded module localized at a generic point of either component has length 1. Concretely:

- Localize at the generic point of $V(\xi)$ (i.e. at the prime (ξ)). In that localization, ξ is nonunitally small and x is a unit, so
$$(\operatorname{gr} M_\alpha)_{(\xi)} \cong \mathbb{C}[x, \xi]_{(\xi)} / (\xi)$$
which is a copy of $\mathbb{C}[x]_{(0)}$ of length 1 over the local ring — so the multiplicity is 1 on the zero section.
- Localize at the generic point of $V(x)$ (i.e. at (x)). There ξ is a unit and
$$(\operatorname{gr} M_\alpha)_{(x)} \cong \mathbb{C}[x, \xi]_{(x)} / (x)$$
which is a copy of $\mathbb{C}[\xi]_{\mathrm{gen}}$ of length 1 — so the multiplicity is 1 on the conormal $T_0^* X$.

Hence both components appear with multiplicity 1.

Therefore the characteristic cycle is

$$\boxed{\mathrm{CC}(M_\alpha) \ = \ [T_X^* X] \ + \ [T_0^* X].}$$

This calculation is independent of α, because the parameter α contributes only lower-order terms that do not affect the principal symbol.

Now consider the de-Rham complex of M_α:

$$\mathrm{DR}(M_\alpha) = [M_\alpha \xrightarrow{\partial_x} M_\alpha].$$

It is exact, this is bad. To fix this, we need to analytify M_α:

$$M_\alpha^{\mathrm{an}} = \mathcal{O}_{X^{\mathrm{an}}} \otimes_{\mathbb{C}[x]} M_\alpha = \mathcal{O}_{X^{\mathrm{an}}}[x^{-1}] \cdot x^\alpha.$$

Example 12.4. $N = \mathbb{C}[\delta]$ where δ is the delta function at 0, $x \cdot \delta = 0$. Then $N = \mathcal{D}_X / \mathcal{D}_X x$. Characteristic cycle is $[T_0^* X]$ with multiplicity 1.

Remark 12.5. The two examples above consists of all regular singularities in some sence.

13. Double Covers and Genuine Representations

2025-10-13

Lemma 13.1. *Let G be a connected Lie group, $\widetilde{G}$ is a double cover. Then the following are equivalent:*

- *$\widetilde{G}$ split;*
- *$\widetilde{G}$ is disconnected;*
- *there exists a genuine representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$.*

Proof. Only need to notice that $\widetilde{G}_0 \to G$ is a covering map of degree 1 or 2. $\square$

There is a same statement for p-adic groups in [Nev99].

Lemma 13.2. *Let G be an arbitrary Lie group, $\pi : \widetilde{G} \to G$ a double cover. Then the following are equivalent:*

- *there exists a genuine finite dimensional representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$;*
- *$z \notin \widetilde{G}_0$;*
- *$z \notin (\pi^{-1}(G_0))_0$;*
- *there exist a genuine finite dimensional representation of $\pi^{-1}(G_0)$ which is trivial on the identity component $(\pi^{-1}(G_0))_0$;*
- *$\pi^{-1}(G_0)$ is disconnected;*
- *$\pi^{-1}(G_0)$ is split.*

Here z is the nontrivial element in $\ker(\pi)$.

If in addition the double cover $\widetilde{G}$ is induced by a square root of a character $\gamma : G \to \mathbb{C}^\times$, i.e. $\widetilde{G} = \{(g, \epsilon) \in G \times \mathbb{C}^\times \mid \epsilon^2 = \gamma(g)\}$, these are also equivalent to:

- *γ has a square root.*

Proof. Only need to notice that $(\pi^{-1}(G_0))_0 = \widetilde{G}_0$, together with results in [Sch87]. $\square$

14. Integrable connections and local systems

2025-10-14

$X = \operatorname{Spec}(R)$ smooth affine with coordinate $x_1, \ldots, x_n$, M is an integrable connection.

Theorem 14.1. *$M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ for some local system L on X^{an}.*

Proof. We know that M is locally free, suppose $M|_U$ is freely generated by $m_1, \cdots, m_n$, suppose $\partial_n \underline{m} = \chi \underline{m}$ where χ is a matrix of holomorphic functions on U. Locally we can expand χ with respect to x_n:

$$\chi = \sum_{i=0}^{\infty} \chi_i(x_1, \ldots, x_{n-1}) x_n^i.$$

Take $\Psi = \sum_{i=1}^{\infty} \frac{1}{i+1} \chi_i(x_1, \ldots, x_{n-1}) x_n^i$, so $\partial_n \Psi = \chi$. Take $\Phi = \exp(-\Psi)$ which is invertible, and set $\underline{s} = \Phi \underline{m}$.

So

$$\partial_n s = \partial_n(\Phi) \cdot \underline{m} + \Phi \cdot (\partial_n \underline{m}) = 0.$$

There is a small open subset $V \subset U$ such that $M^{an}|_V = \bigoplus_{i=1}^{r} \mathcal{O}_V s_i$, take

$$\begin{aligned} M_1 &= \ker(M^{an}|_V \xrightarrow{\partial_n} M^{an}|_V) \\ &= \left\{ \sum_{i=1}^{r} f_i s_i \in M^{an}|_V \;\middle|\; \partial_n(\sum_{i=1}^{r} f_i s_i) = 0, \forall i \right\} \\ &= \left\{ \sum_{i=1}^{r} f_i s_i \in M^{an}|_V \;\middle|\; \partial_n f_i = 0, \forall i \right\} \end{aligned}$$

$$= \bigoplus_{i=1}^{r} \mathcal{O}_{V_1} s_i, \quad V_1 = V \cap \{x_n = 0\}.$$

Continue this process, we can finally get a subsheaf

$$M_n|_V = \bigoplus_{i=1}^{n} \mathbb{C} s_i.$$

of M^{an}, where s_i are horizontal sections. So $M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ where L is the local system generated by s_i.

□

Remark 14.2. The proof actually implies that integrable connections are locally generated by horizontal sections, and the sheaf of horizontal sections is a local system.

Corollary 14.3. $\mathrm{DR}(M) = L$

Proof. $H^0(\mathrm{DR}(M)) \simeq \ker(M \xrightarrow{\nabla} \Omega^1_X \otimes M) = L$. All the other cohomologies vanish because of Poincaré lemma, since this complex is locally de-Rham complex. □

Remark 14.4. This means that de Rham complex of M gives a locally free resolution of L. So the hypercohomology of $\mathrm{DR}(M)$ computes the cohomology of L.

15. Kashiwara's equivalence and its applications

2025-10-16

Kashiwara's equivalence

Y smooth affine $/\mathbb{C}$ with coordinate $x_1, \cdots, x_n, y_1, \cdots, y_l$, $X = \{y_1 = \cdots = y_l = 0\} \subset Y$, X is also smooth. $I = (y_1, \ldots, y_l)$, $\mathcal{D}_Y$ the ring of differential operators on Y, $\mathcal{D}_X$ the ring of differential operators on X. $\mathrm{Mod}_{qc}(\mathcal{D}_X)$: the category of quasi-coherent $\mathcal{D}_X$ modules, $\mathrm{Mod}^X_{qc}(\mathcal{D}_Y)$: the category of quasi-coherent $\mathcal{D}_Y$ modules supported on X.

Theorem 15.1. *(Kashiwara's equivalence) The functor:*

$$i_+ : \mathrm{Mod}_{qc}(\mathcal{D}_X) \to \mathrm{Mod}^X_{qc}(\mathcal{D}_Y), \quad M \mapsto \mathcal{D}_{Y \leftarrow X} \otimes_{\mathcal{D}_X} M = M[\partial_{y_1}, \cdots, \partial_{y_l}],$$

is an equivalence of categories. Its quasi-inverse is given by:

$$i^\sharp : \mathrm{Mod}^X_{qc}(\mathcal{D}_Y) \to \mathrm{Mod}_{qc}(\mathcal{D}_X), \quad N \mapsto \ker(N \longrightarrow^{y} N).$$

Proof. There is a proof in [HTT08], I don't want to repeat it here. □

Remark 15.2. In the case of quasi-coherent sheaves, the equivalence does not hold. For example, Let $Y = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(R/I) \subseteq Y$ a closed subscheme defined by ideal I. Then the functor:

$$i^* : \mathrm{Mod}^X_{qc}(\mathcal{O}_Y) \to \mathrm{Mod}_{qc}(\mathcal{O}_X), \quad N \mapsto R/I \otimes_R N.$$

is not an equivalence of categories. Since if M is an R-module, then M/I^2M is supported on X, but it does not defined over R/I.

Corollary 15.3. *Let M be a quasi-coherent $\mathcal{D}_X$-module, if $Ch(M) = T^*_Y X$ for an irreducible closed smooth subvariety $Y \subset X$, then $M = i_+(N)$ for some integrable connection N on Y.*

$$\begin{tikzcd} \mathrm{Mod}_{qc}(\mathcal{D}_X) \arrow[r, shift left, "i_+"] & \mathrm{Mod}_{qc}^X(\mathcal{D}_Y) \arrow[l, shift left, "i^\sharp"] \end{tikzcd}$$

$$\begin{tikzcd} \boxed{\textbf{Orbital integrals of } G} \arrow[loop above] \arrow[r, dashed, leftrightarrow, "f"'] \arrow[d, dashed, leftrightarrow, "\text{Transfer}"] & \boxed{\textbf{L-packets of } G} \arrow[loop above] \arrow[d, dashed, leftrightarrow, "\text{Endoscopic}"'] \\ \boxed{\textbf{Orbital integrals of } H} \arrow[loop below] \arrow[r, dashed, leftrightarrow, "f_H"'] & \boxed{\textbf{L-packets of } H} \arrow[loop below] \end{tikzcd}$$

16. Nilpotent orbits and degenerate Whittaker models

2025-10-19

Degenerate Whittaker model

G be a finite extension of a reductive algebraic group defined over local field F, $\mathfrak{g}$ be its Lie algebra, fix a nontrivial character $\chi : F \to \mathbb{C}^1$.

A Whittaker pair is a pair (S, φ) where $S \in \mathfrak{g}$ is a rational semi-simple element, $\varphi \in \mathfrak{g}^*$ such that $\mathrm{ad}^*(S)(\varphi) = -2\varphi$. Given such a Whittaker pair, we can define a degenerate Whittaker model $\mathcal{W}_{S,\varphi}$ as follows: Let $\mathfrak{g} = \bigoplus_{r\in\mathbb{Q}} \mathfrak{g}_r^S$ be the eigenspace decomposition with respect to $\mathrm{ad}(S)$, define

$$\mathfrak{u} = \mathfrak{g}_{\geq 1} = \bigoplus_{r\geq 1} \mathfrak{g}_r^S.$$

Define an anti-symmetric form ω_φ on $\mathfrak{u}$ by

$$\omega_\varphi(X, Y) = \varphi([X, Y]), \quad X, Y \in \mathfrak{u}.$$

Let $\mathfrak{n}$ be the kernel of ω_φ, and $\mathfrak{n}' = \mathfrak{n} \cap \ker(\varphi)$. Then $\mathfrak{u}/\mathfrak{n}$ is a symplectic space with symplectic form induced by ω_φ, and $\mathfrak{u}/\mathfrak{n}'$ is a Heisenberg Lie algebra with center $\mathfrak{n}/\mathfrak{n}'$. Corresponding groups are defined by exponentiation:

$$U = \exp(\mathfrak{u}), \quad N = \exp(\mathfrak{n}), \quad N' = \exp(\mathfrak{n}').$$

If $\varphi \neq 0$, there is a unique irreducible representation $(\rho_\varphi, \mathcal{S}_\varphi)$ of U/N' with central character $\chi_\varphi : N/N' \to \mathbb{C}^\times$ defined by

$$\chi_\varphi(\exp(X)) = \chi(\varphi(X)), \quad X \in \mathfrak{n}.$$

Define the degenerate Whittaker model:

$$\mathcal{W}_{S,\varphi} = \mathrm{Ind}_U^G(\rho_\varphi).$$

If $\varphi = 0$, define $\mathcal{W}_{S,0} = \mathrm{Ind}_U^G(1_U)$.

Remark 16.1. In the special case when (S, φ) comes from an $\mathfrak{sl}_2$-triple (e, h, f) with $S = h$ and φ corresponding to f via the Killing form, we can see that $\mathfrak{n} = \mathfrak{g}_{\geq 2}$. Then $\mathcal{W}_f = \mathcal{W}_{S,\varphi}$ is called a generalized Whittaker model associated to the nilpotent orbit of f.

Affine transformation group of real line
$P_2(\mathbb{R}) = \mathbb{R}^\times \ltimes \mathbb{R}$, with multiplication:

$$(a,b)\cdot(a',b') = (aa', b+ab').$$

Its Lie algebra is $\mathfrak{p}_2(\mathbb{R}) = \mathbb{R}\times\mathbb{R}$ with bracket:

$$[(x_1,y_1),(x_2,y_2)] = (0, x_1y_2 - x_2y_1).$$

17. Spectral Sequences of Filtered Complexes

2025-10-21

Spectral Sequences of Filtered Complexes
$(F^\bullet, d)$ be a filtered complex, i.e. a complex $F^\bullet$ with an increasing filtration of subcomplexes:

$$\cdots \subseteq F_j^\bullet \subseteq F_{j+1}^\bullet \subseteq \cdots \subseteq F^\bullet.$$

Such that $d(F_j^i) \subseteq F_j^{i+1}$, and the filtration is exhaustive and separated:

$$\bigcup_{j\in\mathbb{Z}} F_j^i = F^i, \quad \bigcap_{j\in\mathbb{Z}} F_j^i = 0.$$

For any triple (i,j,k), define:

- $Z_j^i(k) := \{u \in F_j^i \mid du \in F_{j-k}^{i+1}\}$;
- $B_j^i(k) := d(F_{j+k-1}^{i-1}) \cap F_j^i$;
- $E_j^i(k) := \frac{Z_j^i(k)+F_{j-1}^i}{B_j^i(k)+F_{j-1}^i}$.

So $d(Z_j^i(k)) \subseteq Z_{j-k}^{i+1}(k)$, which induces a differential $d_k : E_j^i(k) \to E_{j-k}^{i+1}(k)$.

We also define $E_k^{p,q} := E_{-p}^{p+q}(k)$.

Example 17.1. $E_j^i(0) = \frac{F_j^i}{F_{j-1}^i}$, so $E_0^{p,q} = \mathrm{Gr}_{-p}F^{p+q}$.

$E_j^i(1) = H^i(\mathrm{Gr}_j F^\bullet)$, so $E_1^{p,q} = H^{p+q}(\mathrm{Gr}_{-p}F^\bullet)$. The first page of the spectral sequence is just the cohomology of the associated graded complex.

Lemma 17.2. $H^i(E_j^\bullet(k)) = E_j^i(k+1)$.

Definition 17.3. The spectral sequence associated to the filtered complex $(F^\bullet, d)$ is defined to be the collection of complexes $\{(E_\bullet^\bullet(k), d_k)\}_{k\geq 0}$.

They convergent to the cohomology of the original complex: $E_\infty^{p,q} \simeq \mathrm{Gr}_{-p}H^{p+q}(F^\bullet)$, if for any i,j, there exist $k \gg 0$ such that $E_j^i(k) = \mathrm{Gr}_j H^i(F^\bullet)$.

Definition 17.4. If there exist a $w \in \mathbb{Z}_{\geq 0}$ such that for any i,j, $F_j^i \cap d(F^{i-1}) \subseteq d(F_{j+w}^{i-1})$. Then the filtration is called w-regular.

Theorem 17.5. *If $F^\bullet$ is w-regular, then the spectral sequence associated to $(F^\bullet, d)$ converges to the cohomology of the original complex at page $w+1$, i.e. for any i,j, $E_j^i(w+1) \simeq \mathrm{Gr}_j H^i(F^\bullet)$.*

Example 17.6. $K^\bullet$ is a chain complex, F a left exact functor, then there is a natural spectral sequence:

$$E_2^{p,q} = \mathbb{R}^{p+q}F(K^q) \Rightarrow \mathbb{R}^{p+q}F(K^\bullet).$$

(maybe:

$$E_1^{p,q} = \mathbb{R}^p F(K^q) \Rightarrow \mathbb{R}^{p+q} F(K^\bullet)?$$

)

Because we can use the Cartan-Eilenberg resolution to resolve $K^\bullet$, which is given by:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & K^0 \arrow[r] \arrow[d] & I^{0,0} \arrow[r] \arrow[d] & I^{0,1} \arrow[r] \arrow[d] & \cdots \\
0 \arrow[r] & K^1 \arrow[r] \arrow[d] & I^{1,0} \arrow[r] \arrow[d] & I^{1,1} \arrow[r] \arrow[d] & \cdots \\
0 \arrow[r] & K^2 \arrow[r] \arrow[d] & I^{2,0} \arrow[r] \arrow[d] & I^{2,1} \arrow[r] \arrow[d] & \cdots \\
 & \vdots & \vdots & \vdots &
\end{tikzcd}$$

Then we can form a total complex $\mathrm{Tot}(I^{\bullet,\bullet})$ with a natural filtration, and apply the functor F to get a filtered complex $(F(\mathrm{Tot}(I^{\bullet,\bullet})), d)$. The spectral sequence associated to this filtered complex is just the spectral sequence above.

A special case is the Hodge spectral sequence, when X is a smooth algebraic variety over $\mathbb{C}$, $K^\bullet = \Omega_X^\bullet$ the de-Rham complex, and $F = \mathbb{R}\Gamma(X, -)$ the derived global section functor, the spectral sequence is:

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\mathrm{dR}}^{p+q}(X).$$

18. Motivation for Quotient Stacks

2025-10-23

Lemma 18.1. *Let H be a Lie group acting freely and properly on a smooth manifold M, then for any smooth manifold T, there is a natural isomorphism:*

$$\mathrm{Hom}_{sm}(T, M/H) \simeq \frac{\{(P \to^{\pi} T, f : P \to M) | \pi \text{ a principle } H\text{-bundle, } f \text{ is } H\text{-equivariant}\}}{\text{equivalence}}.$$

Proof. Given a smooth map $g : T \to M/H$, we can form the following Cartesian diagram:

$$\begin{tikzcd}
P \arrow[r, "f"] \arrow[d, "\pi"'] & M \arrow[d] \\
T \arrow[r, "g"'] & M/H
\end{tikzcd}$$

Since H acts freely and properly on M, $\pi : P \to T$ is a principle H-bundle, and $f : P \to M$ is H-equivariant.

Conversely, given a principle H-bundle $\pi : P \to T$ and an H-equivariant map $f : P \to M$, we can define a smooth map $g : T \to M/H$ by sending $t \in T$ to the orbit of $f(p)$ where $p \in \pi^{-1}(t)$. $\square$

Remark 18.2. Algebraic geometers use this to define the quotient stack $[M/H]$ even if the action is not free or proper, but $\mathrm{Hom}(T, [M/H])$ need to be a groupoid, not just a set of equivalence classes.

19. Unipotent Arthur Packets [ABV92]

2025-10-29

Let G be a connected reductive algebraic group over $\mathbb{R}$, fix a weak E-group $\check{G}^{\Gamma}$. Which is an extension:

$$1 \to \check{G} \to \check{G}^{\Gamma} \to \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \to 1.$$

Here $\check{G}$ is the complex dual group of G. An L-parameter is a continuous homomorphism:

$$\phi : W_{\mathbb{R}} \to \check{G}^{\Gamma},$$

such that the composition $W_{\mathbb{R}} \to {}^L\check{G} \to \mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is the natural projection and $\phi(\mathbb{C}^*)$ consists of semisimple elements of $\check{G}$.

Given an L-parameter ϕ, we can attach the following data:

- two semisimple elements $\lambda, \mu \in \check{G}$, defined by $\phi(e^t) = \exp(\lambda t + \mu \bar{t})$;
- $y = \exp(\pi i \lambda)\phi(j) \in \check{G}^{\Gamma} - \check{G}$.

The L-parameter ϕ is called *tempered* if the closure of $\phi(W_{\mathbb{R}})$ in the complex analytic topology is compact. This is equivalent to say that $\lambda + \mathrm{Ad}(y)\lambda$ is an elliptic element of $\check{\mathfrak{g}}$.

Fix a geometric parameter (y, Λ), it lies in a variety $X_y(\mathcal{O}, \check{G}^{\Gamma}) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$. The $\check{G}$-equivariant microlocal geometry on $X_y(\mathcal{O}, \check{G}^{\Gamma})$ is in some sense equivalent to the $K(y)$-equivariant microlocal geometry on $\check{G}(\lambda)_0/P(\Lambda)$.

Roughly speaking, there is a bijection between:

$$\boxed{\text{representation with minimal GK-dim}}$$
$$\updownarrow$$
$$\boxed{\text{parameters supported on regular orbits in } \check{G}(\Lambda)_0/P(\Lambda)}$$

Definition 19.1. An Arthur parameter is a homomorphism:

$$\psi : W_{\mathbb{R}} \times SL(2, \mathbb{C}) \to \check{G}^{\Gamma},$$

such that the restriction to $W_{\mathbb{R}}$ (ψ_0) is a tempered L-parameter, and the restriction to $SL(2, \mathbb{C})$ (ψ_1) is algebraic.

For an arthur parameter ψ, we can define an L-parameter ϕ_ψ by:

$$\phi_\psi(w) = \psi(w, \begin{pmatrix} |w|^{1/2} & 0 \\ 0 & |w|^{-1/2} \end{pmatrix}).$$

Let $y_1 = \psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \check{G}$, and $\lambda_1 = d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \in \check{\mathfrak{g}}$, (y_0, λ_0) is the parameter defined by the restriction of ψ to $W_{\mathbb{R}}$.

Then we have $y = y_0 y_1, \lambda = \lambda_0 + \lambda_1$, since $\phi_\psi(e^t) = \exp(\lambda_0 t + \mu_0 \bar{t})\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix}$, and

$$\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix} = \exp\left(t \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \bar{t} \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right) = \exp(\lambda_1 t + \lambda_1 \bar{t}).$$

So $\phi_\psi(e^t) = \exp((\lambda_0 + \lambda_1)t + (\mu_0 + \lambda_1)\bar{t})$, hence $\lambda = \lambda_0 + \lambda_1$.

And

$$\begin{aligned} y &= \exp(\pi i \lambda)\phi_\psi(j) \\ &= \exp(\pi i \lambda_0) \exp\left(\pi i d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right) \psi_0(j) \\ &= \exp(\pi i \lambda_0)\psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \psi_0(j) \\ &= y_0 y_1. \end{aligned}$$

In addition of the two semisimple elements λ, μ and y defined as above, we can also define a nilpotent element $E_\psi \in \check{\mathfrak{g}}$ by:

$$E_\psi = d\psi_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \check{\mathfrak{g}}.$$

E_ψ enjoys the following properties:

- $\mathrm{Ad}(y)E_\psi = -E_\psi$, this means $E_\psi \in \mathfrak{s}(y)$;
- $[\lambda, E_\psi] = E_\psi$, in other words, $E_\psi \in \check{\mathfrak{g}}(\Lambda)_1 \subseteq \mathfrak{n}(\lambda)$ where $\check{\mathfrak{g}}(\Lambda) = \bigoplus_{r \in \mathbb{Z}} \check{\mathfrak{g}}_r$ is the eigenspace decomposition with respect to $\mathrm{ad}(\lambda)$.

$$\begin{gathered} \text{Langlands parameter } (y, \lambda) \\ \rightsquigarrow \\ S = \check{G} \times^{K(y)} eP(\Lambda) \subseteq X_y(\mathcal{O}, \check{G}^\Gamma) = \check{G} \times^{K(y)} \check{G}(\Lambda)/P(\Lambda) \\ \rightsquigarrow \\ S_K = K(y) \cdot (eP(\Lambda)) \subseteq \check{G}(\Lambda)_0/P(\Lambda) \end{gathered}$$

Cotangent space at the base point:

$$T^*_{eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{n}(\lambda).$$

Conormal space at the base point of the $K(y)$-orbit:

$$T^*_{K(y), eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{s}(y) \cap \mathfrak{n}(\lambda).$$

So we can view E_ψ as an element in the conormal space $T^*_{K(y), eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda))$ also in $T^*_{\check{G}, \phi_\psi}(X(\check{G}^\Gamma))$.

Definition 19.2. An Arthur parameter ψ is called *unipotent* if the restriction to $\mathbb{C}^* \subseteq W_\mathbb{R}$ is trivial.

Remark 19.3. In this case, $\psi : \mathrm{SL}_2(\mathbb{C}) \times \Gamma \to \check{G}^\Gamma$. So $\lambda = \lambda_1$

Lemma 19.4. *Suppose ψ is unipotent. Consider the partial Springer resolution*

$$\mu : T^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \check{G}(\Lambda)_0 \times^{P(\Lambda)} \mathfrak{n}(\lambda) \to \mathcal{N}_{\mathcal{P}(\Lambda)},$$

$E_\psi \in \mathcal{N}_{\mathcal{P}(\Lambda)} \cap \mathfrak{s}(y)$ is a regular element (belong to the Richardson orbit), and the fiber $\mu^{-1}(E_\psi)$ is a single point. This guarantees that μ has degree 1.

Proof. We can see that

$$\mathfrak{p}(\Lambda) = \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^2} \cap \mathrm{Ad}(E_\psi)\check{\mathfrak{g}} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^3} \cap \mathrm{Ad}(E_\psi)^2\check{\mathfrak{g}} + \cdots,$$

So the centralizer $\check{G}(\Lambda)^{E_\psi} \subseteq P(\Lambda)$, which means that the fiber $\mu^{-1}(E_\psi)$ is a single point. The operator E_ψ carry the sum of non-negative eigenspaces of $\mathrm{ad}(\lambda)$ onto the sum of positive eigenspaces, i.e. $\mathrm{Ad}(E_\psi)\mathfrak{p}(\Lambda) = \mathfrak{n}(\Lambda)$. In other words, the differential of the map:

$$P(\Lambda) \to \mathfrak{n}(\lambda), \quad p \mapsto \mathrm{Ad}(p)E_\psi,$$

is surjective at the identity coset, so the orbit $\check{G}(\Lambda) \cdot E_\psi$ is dense in $\mathfrak{n}(\lambda)$, which means that E_ψ is a regular element in $\mathcal{N}_{\mathcal{P}(\Lambda)}$. $\square$

Theorem 19.5. *Fix an algebraic morphism $\psi : \mathrm{SL}_2(\mathbb{C}) \to \check{G}^\Gamma$, let $\lambda = d\psi \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$, $\mathcal{O} = \check{G} \cdot \lambda \subseteq \check{\mathfrak{g}}$ the corresponding semisimple orbits. Consider the open and closed subvariety of the geometric parameter space $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$.*

Then there are bijections between:

- *equivalence classes of unipotent Arthur parameters supported on $X_y(\mathcal{O}, \check{G}^\Gamma)$;*
- *regular $K(y)$-orbits in $\check{G}(\Lambda)_0/P(\Lambda)$;*
- *regular $K(y)$-orbits in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.*

Proof. Given a unipotent Arthur parameter ψ, we can associate to it a nilpotent element $E_\psi \in \mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ as above, then $K(y) \cdot E_\psi$ is a regular orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.

Conversely, given a regular $K(y)$-orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$, let e be a representative. We can construct a homomorphism $\psi_1' : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ such that $d\psi_1' \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = e$, and

$$\psi_1'(\theta_{\mathrm{SL}_2}(x)) = \theta_y(\psi_1'(x)).$$

Define $y_0 = y\psi_1' \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$

Then (y_0, ψ_1') is the corresponding unipotent Arthur parameter. $\square$

Remark 19.6. In other words, this theorem tell us the following diagram:

$$\begin{tikzcd}
\boxed{\check{G}\text{-orbits in } \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))} \arrow[r, leftrightarrow, "1-1"] & \boxed{K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \\
\boxed{\text{unipotent A-parameters supported on } X_y(\mathcal{O}, \check{G}^\Gamma)} \arrow[u, hook] \arrow[r, leftrightarrow, "1-1"] & \boxed{\text{regular } K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \arrow[u, hook]
\end{tikzcd}$$

Theorem 19.7. *Under the above setting, fix a unipotent Arthur parameter ψ supported on $X_y(\mathcal{O}, \check{G}^\Gamma)$, let S be the corresponding $\check{G}$-orbit in $X_y(\mathcal{O}, \check{G}^\Gamma)$, S_K be the corresponding $K(y)$-orbit in $\check{G}(\Lambda)_0/P(\Lambda)$, and $\mathcal{O}_K$ be the corresponding $K(y)$-orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.*

Then there are bijections between:

- *irreducible $(\check{\mathfrak{g}}(\Lambda), K(y)^{alg})$-modules M annihilated by $I_{\mathcal{P}(\Lambda)}$ such that*
$$\overline{\mathcal{O}_K} \subseteq \mathrm{AV}(M);$$
- *irreducible $K(y)^{alg}$-equivariant $\mathcal{D}$-modules $\mathcal{M}$ on $\check{G}(\Lambda)_0/P(\Lambda)$ such that*
$$T^*_{S_K}(\check{G}(\Lambda)_0/P(\Lambda)) \subseteq \mathrm{Ch}(\mathcal{M});$$
- *irreducible $\check{G}$-equivariant $\mathcal{D}$-modules $\mathcal{N}$ on $X_y(\mathcal{O}, \check{G}^\Gamma)$ such that*
$$T^*_S(X_y(\mathcal{O}, \check{G}^\Gamma)) \subseteq \mathrm{Ch}(\mathcal{N});$$
- *Arthur packets $\Pi^{Art}_\psi(G/\mathbb{R})$ associated to ψ.*

Corollary 19.8. *Fix a nilpotent orbit $\check{\mathcal{O}} \subseteq \check{\mathfrak{g}}$, attached to it is a semisimple orbit $\check{\mathcal{O}}_{ss} \subseteq \check{\mathfrak{g}}$ then*
$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \bigcup_{\psi \text{ unipotent}, \mathcal{O}_\psi = \check{\mathcal{O}}_{ss}} \Pi^{Art}_\psi(G/\mathbb{R}).$$

Remark 19.9.
$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \{\text{irreducible representations in } \mathrm{Irr}_{\check{\mathcal{O}}_{ss}}(G/\mathbb{R}) \text{ with minimal GK-dim}\}$$

Example 19.10. $G = G_1 \times G_1$ where G_1 is a complex connected reductive algebraic group viewed as a real group. Then $\check{G}^\Gamma = (\check{G}_1 \times \check{G}_1) \rtimes \{1, \delta\}$. A homomorphism $\psi : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ is determined by two homomorphisms $\psi_1, \psi_2 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$, we can check that ψ comes from a unipotent Arthur parameter if and only if ψ_1 and ψ_2 are conjugate to each other. So unipotent Arthur parameters of G are in bijection with homomorphisms $\psi_1 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$. Therefore, equivalence classes of unipotent Arthur parameters of G_1 are in bijection with nilpotent orbits in $\check{\mathfrak{g}}_1$.

20. Questions on the Unitary Dual

2025-11-04

Some questions about unitary dual problem

- How can we reduce the classification of irreducible unitary $(\mathfrak{g}, K)$-module to the classification of such module with real infinitesimal character?
- Why there always exist a $\mathfrak{u}_\mathbb{R}$-invariant Hermitian form on irreducible $(\mathfrak{g}, K)$-module with real infinitesimal character?
- what is Jantzen filtration of a standard $(\mathfrak{g}, K)$-module? Why it coincides with Hodge filtration?

Let G be a connected reductive group over $\mathbb{C}$, $\mathcal{B}$ the variety of all Borel subalgebras of $\mathfrak{g}$, ${}^a\mathfrak{h}$ is the universal Cartan subalgebra. For any Borel subalgebra $\mathfrak{b} \in \mathcal{B}$, we have a canonical isomorphism:
$$\phi_\mathfrak{b} : {}^a\mathfrak{h} \simeq \mathfrak{h},$$
such that negative roots in ${}^a\mathfrak{h}^*$ correspond to roots in $\Delta(\mathfrak{h}, \mathfrak{n})$.

We can define a G-equivariant vector bundle $\mathcal{L}_\lambda$ over $\mathcal{B}$ for any $\lambda \in {}^a\mathfrak{h}^*$ by:

$$\mathcal{L}_\lambda|_{\mathfrak{b}} = \mathbb{C}_\lambda := \text{1-dimensional representation of } B \text{ defined by } \lambda \circ \phi_{\mathfrak{b}}^{-1}.$$

$$\boxed{\text{Weight lattice } \Lambda \subseteq {}^a h^*} \xleftrightarrow{1-1} \boxed{G\text{-equivariant line bundles on } \mathcal{B}}$$

Under this correspondence, $\mathcal{L}_\lambda$ is ample if and only if λ is regular dominant. $\mathcal{L}_{-2\rho}$ is the canonical line bundle $\wedge^{top} T^*\mathcal{B}$ on $\mathcal{B}$.

21. Annihilator Varieties via Cells

2025-11-19

21.1. compute annihilator variety.

$$\boxed{\mathrm{Prim}_\nu(\mathrm{U}(\mathfrak{g}))} \xleftrightarrow{1-1} \boxed{\text{left cells in } \mathrm{Coh}_\nu(\mathfrak{g}, \mathfrak{b})} \longrightarrow \boxed{\text{double cells in } \mathrm{Coh}_\nu(\mathfrak{g}, \mathfrak{b})}$$

$$I \in \mathrm{Prim}(\mathrm{U}(\mathfrak{g})) \longmapsto \mathcal{C}_{\nu,I}^{L} \longmapsto \mathcal{C}_{\nu,I}^{LR}$$

$\mathrm{span}(\mathcal{C}_{\nu,I}^{LR})$ is a representation of the integral Weyl group $W(\nu)$, all irreducible constituents of $\mathrm{span}(\mathcal{C}_{\nu,I}^{LR})$ lies in a single double cell, and contains a unique special representation of $W(\nu)$

$$\sigma_{\nu,I}.$$

Then by results of Joseph, we have:

$$\mathrm{AV}(I) = \mathcal{O}(j_{W(\nu)}^{W}(\sigma_{\nu,I})).$$

Which means that to compute the annihilator variety of $\mathrm{L}(w\nu)$, we only need to compute the corresponding cell representation.

In the case of $(\mathfrak{g}, K)$-modules, the method is the same:

$$\begin{array}{c} X\colon \text{Irreducible } (\mathfrak{g}, K)\text{-module} \\ \rightsquigarrow \\ C\colon \text{Harish-Chandra cell containing } X \\ \rightsquigarrow \\ \sigma_C\colon \text{special representation of } W(\nu) \text{ contained in the cell representation} \\ \rightsquigarrow \\ \mathcal{O}(j_{W(\nu)}^{W}(\sigma_C)) = \mathrm{AV}(X) \end{array}$$

Remark 21.1. The key point is the result of Joseph [Jos85] the Springer correspondence of cell representation gives the annihilator variety. However, I'm not fully understand the details.

21.2. **annihilator varieties are special.** j-induction of parabolic subgroups preserves special representations.

To prove that the annihilator variety of irreducible $(\mathfrak{g}, K)$-module is closure of a single special nilpotent orbit, only need to check the case when infinitesimal character has good parity and bad parity.

For example, in the case of $G = \mathrm{Sp}_{2n}(\mathbb{R})$, we have:

$$\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b, n_b)) \otimes \mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R})) \simeq \mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2n}(\mathbb{R})).$$

We need to show that the j-induction of any special repn appears in $\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b, n_b))$ and any special repn appears in $\mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R}))$ gives a special repn in $\mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2n}(\mathbb{R}))$.

- Problem: How to compute the j-induction explicitly?
- Problem: the special repn of each cell always appears?

22. Annihilator and Whittaker Support Problems

2025-11-21

Some problems/plans

- Use counting methods to prove that all annihilator variety of irreducible $(\mathfrak{g}, K)$-modules of orthogonal or special groups are special.
- Use counting methods to prove that all annihilator variety of irreducible genuine $(\mathfrak{g}, K)$-modules of metaplectic groups are metaplectic special.
- By computation, prove that in the case of metaplectic groups, metaplectic quasi-admissible equivalent to metaplectic special. solved! Yes
- Prove that for metaplectic groups, whittaker support of irreducible genuine $(\mathfrak{g}, K)$-modules are metaplectic quasi-admissible. Use arguments of Gomez-Gourevitch-Sahi
- Prove associated cycles coincides with whittaker cycles for irreducible $(\mathfrak{g}, K)$-modules of real classical groups and metaplectic groups.

23. Robinson-Schensted and Annihilator Varieties

2025-11-24

For a Weyl group W, elements of W are participated into double cells, left cells, etc. Each double cell $\mathcal{C}^{LR}$ contains a unique special representation $\sigma_{\mathcal{C}^{LR}}$ of W. And each left cell $\mathcal{C}^L$ corresponds to a primitive ideal $I_{\mathcal{C}^L}$ of $\mathrm{U}(\mathfrak{g})$ with infinitesimal character ρ. For each element $w \in W$, there are two processes to get a nilpotent orbit $\mathcal{O}_w$ in $\mathfrak{g}^*$:

$$w \longmapsto \mathcal{C}_w^{LR} \longmapsto \sigma_{\mathcal{C}_w^{LR}} \longmapsto \mathcal{O}(\sigma_{\mathcal{C}_w^{LR}});$$

$$w \xmapsto{RS} P(w) \longmapsto \mathcal{O}_{P(w)} \longmapsto \mathcal{O}_{P(w),s}.$$

The question is: are these two nilpotent orbits the same? i.e. $\mathcal{O}(\sigma_{\mathcal{C}_w^{LR}}) = \mathcal{O}_{P(w),s}$?
answer: Yes! [BV82]

24. Linear Double Covers and Genuine Representations

2026-03-03

G is a linear real reductive group, $\widetilde{G}$ is a double cover of G.

Theorem 24.1. *$\widetilde{G}_0$ is a linear group (i.e. there is an injective homomorphism $\widetilde{G}_0 \to \mathrm{GL}_n(\mathbb{R})$ for some n) equivalent to the existence of a finite-dimensional genuine representation of $\widetilde{G}$.*

Proof. If $\widetilde{G}_0$ is a linear group, there are two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\{1, \varepsilon\} \times \widetilde{G}_0$ is a subgroup of $\widetilde{G}$ with finite index, and it has a finite-dimensional representation such that ε acts by -1, so we get a finite-dimensional genuine representation of $\widetilde{G}$ by induction.
- $\varepsilon \in \widetilde{G}_0$: denote $\widetilde{G}_0$ by H, then $H = A \times {}^0H$ as a Nash group, and ${}^0H$ is also a linear group. So there is a closed embedding of Nash groups ${}^0H \hookrightarrow \mathrm{GL}_n(\mathbb{R})$ for some n (${}^0H$ has compact center), take Zariski closure of ${}^0H$ in $\mathrm{GL}_n(\mathbb{C})$ we get the complexification ${}^0H_{\mathbb{C}}$ of ${}^0H$, there is an algebraic matrix coefficient of ${}^0H_{\mathbb{C}}$ such that ε acts on it by -1, and it generates a finite-dimensional genuine representation of $\widetilde{G}$ by induction.

Conversely, if there is a finite-dimensional genuine representation of $\widetilde{G}$, we also have two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\widetilde{G}_0 \to G$ is injective, since G is a linear group, $\widetilde{G}_0$ is also a linear group.
- $\varepsilon \in \widetilde{G}_0$: then we can find a finite-dimensional representation E_1 of $\widetilde{G}_0$ such that ε acts by -1. We also have a finite-dimensional faithful representation E_2 of G, we can regard it as a representation of $\widetilde{G}_0$ by pullback, and ε acts trivially on E_2. So $E_1 \oplus E_2$ is a finite-dimensional faithful representation of $\widetilde{G}_0$.

□

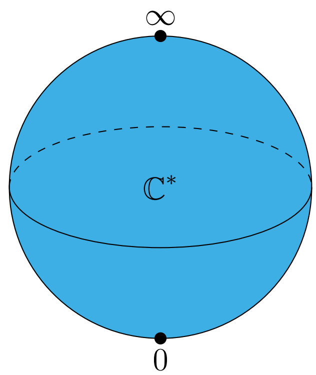
Remark 24.2. After I have a sleep, I find that $\widetilde{G}_0$ can be replaced by $\widetilde{G}$ in this theorem, and the proof is even easier.

25. Geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$

2026-03-21

In this section, we illustrate the geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$, which serve as a simple example for understanding more general phenomena.

$K = \mathbb{C}^*$ action has three orbits on $\mathbb{P}^1(\mathbb{C})$: two closed orbits $\{0\}$ and $\{\infty\}$, and one open orbit $\mathbb{C}^*$. The conormal bundle of the open orbit is the zero section, while the conormal bundle of the closed orbits are the cotangent fiber at 0 and ∞ respectively. So we have three conormal bundles, which are all Lagrangian subvarieties in $T^*\mathbb{P}^1(\mathbb{C})$.

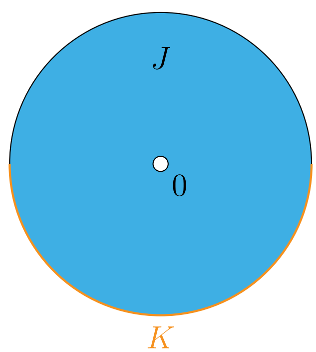


For a K-equivariant perverse sheaf $\mathcal{F}^\bullet$ on it, we calculate its characteristic cycle $\mathrm{CC}(\mathcal{F}^\bullet)$ use the formula

$$\chi_S^{mic} = \sum_{S' > S} c(S, S')\chi_{S'}^{loc},$$

only need to determine the coefficients $c(S, S')(= \sum_i (-1)^i H_c^i(J \cap S', K \cap S'; \mathbb{C}))$ for $S' > S$, where (J, K) is the normal Morse datum of S, which is well-defined up to homeomorphism.

- For the open orbit $\mathbb{C}^*$, we have $S' = \mathbb{C}^*$ and $c(\mathbb{C}^*, \mathbb{C}^*) = -1$.
- For the closed orbit $\{0\}$, we have $S' = \{0\}, \mathbb{C}^*$, and $c(\{0\}, \{0\}) = 1$, for $c(\{0\}, \mathbb{C}^*)$, the pair $(J \cap S', K \cap S')$ is the following picture



therefore $c(\{0\}, \mathbb{C}^*) = \sum_i (-1)^i H_c^i(J \cap \mathbb{C}^*, K \cap \mathbb{C}^*; \mathbb{C}) \overset{?}{=} -1$.

So we have

$$\begin{aligned}
\chi_0^{mic} &= \chi_0^{loc} - \chi_{\mathbb{C}^*}^{loc}, \\
\chi_\infty^{mic} &= \chi_\infty^{loc} - \chi_{\mathbb{C}^*}^{loc}, \\
\chi_{\mathbb{C}^*}^{mic} &= -\chi_{\mathbb{C}^*}^{loc}.
\end{aligned}$$

26. Perverse sheaves on $\mathbb{C}$

2026-03-22

Constructible sheaves on $\mathbb{C} = \mathbb{C}^* \sqcup \{0\}$. Some calculations comes from the lecture notes by Geordie Williamson [Wil15].

Denote $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ the open inclusion, $i : \{0\} \hookrightarrow \mathbb{C}$ the closed inclusion. Then we have the following six functors:

$$\begin{tikzcd} D^b(\mathbb{C}^*) \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & D^b_c(\mathbb{C}) \arrow[l, "j^*=j^!" description] \arrow[r, shift left=2, "i^*"] \arrow[r, shift right=2, "i^!"'] & D^b(\{0\}) \arrow[l, "i_*=i_!" description] \end{tikzcd}$$

$\mathcal{L}$ is a local system on $\mathbb{C}^*$.

$j_!\mathcal{L}$ has no quotient supported on $\{0\}$, j_* has no subobject supported on $\{0\}$. $j_!\mathcal{L}[1]$ and $j_*\mathcal{L}[1]$ are perverse sheaves on $\mathbb{C}$.

Remark 26.1. Actually when $j : S \hookrightarrow X$ is affine locally closed embedding, $j_!$ and j_* are t-exact for the perverse t-structure, so $j_!\mathcal{L}[\dim S]$ and $j_*\mathcal{L}[\dim S]$ are perverse sheaves on X.

By easy calculation of group cohomology, we have:

$$H^i(j_*\mathcal{L}[1]_0) = \begin{cases} 0 & i \neq 0, -1 \\ V_\mu & i = 0 \\ V^\mu & i = -1 \end{cases}$$

There is an exact sequence of perverse sheaves on $\mathbb{C}$:

$$0 \to i_*V^\mu \to j_!\mathcal{L}[1] \to j_*\mathcal{L}[1] \to i_*V_\mu \to 0.$$

27. Enhanced flag variety and line bundles on $\mathcal{B}$

2026-03-31

I. The Enhanced Flag Variety and G_0/N Isomorphism [BB93]

$\mathfrak{g}$ be a complex semisimple Lie algebra, ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}]$ be the universal Cartan subalgebra, G be the automorphism group of $\mathfrak{g}$, G_0 be the connected component of G, it is the adjoint group of $\mathfrak{g}$.

Attached to ${}^a\mathfrak{h}$ is a root system Δ, and the Weyl group W. Although there isn't a well defined action of ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}]$ on $\mathfrak{g}/\mathfrak{b} = \mathfrak{n}^*$, the action of ${}^a\mathfrak{h}$ on $(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^*$ is well defined, and there is a canonical decomposition:

$$(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^* = \bigoplus_{\alpha \in \Sigma} (\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^*_\alpha,$$

Where Σ is the set of simple roots. Note that this choice of positive root system is opposite to the one usually used in representation theory, but it is more natural in geometry since dominant implies positive line bundle in this way.

The enhanced flag variety $\widetilde{\mathcal{B}}$ is defined as the variety of pairs $(\mathfrak{b}, \{x_\alpha\}_{\alpha \in \Sigma})$ where $\mathfrak{b}$ is a Borel subalgebra of $\mathfrak{g}$, and ξ is a nonzero element in $(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^*_\alpha$. There is a natural projection map $\pi : \widetilde{\mathcal{B}} \to \mathcal{B}$ which forgets the second component, and is an H principal bundle.

There is a G action on $\widetilde{\mathcal{B}}$ defined by $g \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha \in \Sigma}) = (\mathrm{Ad}(g)\mathfrak{b}, \{\mathrm{Ad}^*(g)x_\alpha\}_{\alpha \in \Sigma})$. There is also an H action on $\widetilde{\mathcal{B}}$ defined by $h \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha \in \Sigma}) = (\mathfrak{b}, \{\alpha(h)x_\alpha\}_{\alpha \in \Sigma})$. These two actions is compatible in the following way:

$$g \cdot h \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha \in \Sigma}) = \kappa_g(h) \cdot g \cdot (\mathfrak{b}, \{x_\alpha\}_{\alpha \in \Sigma}),$$

where $\kappa : G \to \mathrm{Aut}(H)$ is the action as outer automorphisms of H.

The adjoint group G_0 acts transitively on $\widetilde{\mathcal{B}}$, and the stabilizer of a base point $(\mathfrak{b}, \{x_\alpha\}_{\alpha\in\Sigma})$ is N. So we have an isomorphism $\widetilde{\mathcal{B}} \simeq G_0/N$. Under this identification, the G action on $\widetilde{\mathcal{B}}$ is given by $g\cdot(g_0N) = gg_0(N)$, and the H action on $\widetilde{\mathcal{B}}$ is given by $h\cdot(g_0N) = g_0hN$.

II. Line Bundles on $\mathcal{B}$ and Very Ampleness

Now G is a connected reductive group.

Lemma 27.1. *Let $G \to \mathrm{GL}(V)$ be a finite-dimensional representation of G, $k\cdot v \subseteq V$ be a one-dimensional subspace, $P = \mathrm{Stab}_G(k\cdot v)$. Suppose that k_λ be the one-dimensional representation of P defined by the action of P on $k\cdot v$, then $i^*(\mathcal{O}_{\mathbb{P}(V)}(1)) = \mathcal{O}_{G/P}(\lambda)$.*

Proof. Note that $\mathcal{O}_{\mathbb{P}(V)}(-1)$ is the tautological line bundle, P action on the fiber of $\mathcal{O}_{\mathbb{P}(V)}(-1)$ at $k\cdot v$ is given by k_λ. So $k_{-\lambda}$ for $\mathcal{O}(1)$, also note that $\mathcal{O}_{G/P}(\lambda) = G\times^P k_{-\lambda}$. □

Lemma 27.2. *$\mathcal{O}_{\mathcal{B}}(\lambda)$ is very ample if and only if λ is regular dominant ($\lambda \in X^+(H)$).*

Proof. Consider the Verma module $V^\lambda = \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{h})} k_\lambda$. By the transitive group action of G on $\mathcal{B}$, $\mathcal{O}(\lambda)$ is generated by global sections $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) = V^\lambda$. Hence there is a surjection $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) \twoheadrightarrow \mathcal{O}(\lambda)\otimes_k k_{eB} = k_\lambda$. Taking dual, we have $k_{-\lambda} \hookrightarrow (V^\lambda)^*$.

Since λ is regular dominant, the stablizer of $k_{-\lambda}$ in $(V^\lambda)^*$ is the Borel B, by the previous lemma, we have a G-equivariant morphism $\mathcal{B} \hookrightarrow \mathbb{P}((V^\lambda)^*)$ such that $\mathcal{O}(\lambda)$ is the pullback of $\mathcal{O}(1)$, hence $\mathcal{O}(\lambda)$ is very ample. □

28. Tangent Maps for Associated Bundles

2026-04-05

G is a complex algebraic group, P is a closed subgroup of G with a finite dimensional rational representation V. suppose that $\mathfrak{g} = \mathfrak{u}\oplus\mathfrak{p}$.

$$\begin{array}{ccc} Y = G\times^P V & \xrightarrow{\pi} & V \\ \downarrow & & \\ G/P & & \end{array}$$

There is an isomorphism:

$$\mathfrak{u}\oplus V \xrightarrow{\sim} T_{\overline{(1,\alpha)}}(G\times^P V),$$

which sends (u,v) to $u^\sharp + v$, where $v\in V$ is regarded as a tangent vector of the fiber V at α, and $u^\sharp$ is the vector field on Y generated by the action of $u \in \mathfrak{u} \subset \mathfrak{g}$.

The tangent map of π at $\overline{(1,\alpha)}$ is:

$$d\pi_{\overline{(1,\alpha)}} : \mathfrak{u}\oplus V \to T_{\overline{(1,\alpha)}}(G\times^P V) \to V,$$
$$d\pi_{\overline{(1,\alpha)}}(u,v) = u\cdot\alpha + v,$$

since $\pi(g,\alpha) = g\cdot\alpha$.

The above calculation is instrumental in extending the Kostant-Kirillov symplectic form of nilpotent orbits to the Jacobson-Morozov resolution of a nilpotent orbit:

$$\begin{tikzcd} G \times^P \mathfrak{g}_{\geq 2} \arrow[r, "\pi"] & \overline{\mathbb{O}} \\ G \times^P P \cdot e = G/Z_P(e) \arrow[u, hook] \arrow[r] & \mathbb{O} \arrow[u, hook] \end{tikzcd}$$

$$\pi^* \omega_{KK,\alpha}((u,v),(x,y)) = \kappa(\alpha,[u,x]) + \kappa(u,y) - \kappa(v,x).$$

Where $(u,v),(x,y) \in \mathfrak{g}_{\leq -1} \oplus \mathfrak{g}_{\geq 2} = T_{\overline{(1,\alpha)}}(G \times^P \mathfrak{g}_{\geq 2})$.

This 2-form can be extended to a G-invariant 2-form ω_Y on $G \times^P \mathfrak{g}_{\geq 2}$.

It is easy to check that $\mathrm{Rad}(\omega_Y) = \mathfrak{g}_{-1} \oplus 0$, so this form is symplectic if and only if $\mathfrak{g}_{-1} = 0$, which is equivalent to the nilpotent orbit $\mathbb{O}$ is even.

29. Symplectic Geometry and Formal Quantization

2026-04-22

[Los22]

29.1. **Examples of Symplectic Form and Moment map.**

Proposition 29.1. *The Kostant-Kirillov form ω_{KK} on a coadjoint orbit $\mathbb{O}$ is closed and hence symplectic.*

Proof. For $X \in \mathfrak{g}$ we have a vector field $X^\sharp$ on $\mathbb{O}$ defined by

$$\boxed{X^\sharp_\alpha = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathrm{Ad}^*(\exp(tX))\alpha = [X,\alpha]}$$

Define a smooth function $f_X = \kappa(-,X)$ on $\mathbb{O}$, then we have

$$\mathrm{d}f_X(Y^\sharp) = \kappa([Y,\alpha],X) = \kappa(\alpha,[X,Y]) = \omega_{KK}(X^\sharp,Y^\sharp).$$

So $\iota_{X^\sharp}\omega_{KK} = df_X$.

By Cartan's magic formula, we have

$$\mathcal{L}_{X^\sharp}\omega_{KK} = \mathrm{d}\iota_{X^\sharp}\omega_{KK} + \iota_{X^\sharp}\mathrm{d}\omega_{KK} = \mathrm{dd}f_X + \iota_{X^\sharp}\mathrm{d}\omega_{KK} = \iota_{X^\sharp}\mathrm{d}\omega_{KK}.$$

Since ω_{KK} is G-invariant, $\mathcal{L}_{X^\sharp}\omega_{KK} = 0$, hence $\iota_{X^\sharp}\mathrm{d}\omega_{KK} = 0$ for all $X \in \mathfrak{g}$. Since the vector fields $X^\sharp$ span the tangent space of $\mathbb{O}$ at each point, we have $\mathrm{d}\omega_{KK} = 0$. ∎

Now, we consider the simplest example of Hamiltonian G-space.

Let (V,ω) be a complex symplectic space, and $G = \mathrm{Sp}(V,\omega)$

Proposition 29.2. *The moment map $\mu : V \to \mathfrak{g}^*$ is given by $(\mu(v),X) = \frac{1}{2}\omega(Xv,v)$ for $v \in V$ and $X \in \mathfrak{g}$.*

Proof. Set $H_X(v) = (\mu(v),X)$, we only need to check that $\iota_{X^\sharp}\omega = \mathrm{d}H_X$:

$$\begin{aligned}(\mathrm{d}H_X)_v(\delta v) &= \frac{1}{2}\omega(X\delta v,v) + \frac{1}{2}\omega(Xv,\delta v) \\ &= \omega(Xv,\delta v) \\ &= (\iota_{X^\sharp}\omega)_v(\delta v).\end{aligned}$$

Where δv is a tangent vector of V at v, which can be identified with an element in V itself. $\square$

Remark 29.3. Note that $X^\sharp(v) = X \cdot v$ for $v \in V$.

Lemma 29.4. *The symplectic group G acts transitively on $V \setminus \{0\}$.*

Proof. Use induction on the dimension. $\square$

Proposition 29.5. *The moment map $\mu : V \setminus \{0\} \to \mathfrak{g}^*$ is a double cover onto the minimal nilpotent orbit $\mathbb{O}_{min}$ in $\mathfrak{g}^*$.*

Proof. Use trace form on $\mathfrak{g}$ to identify $\mathfrak{g}$ with $\mathfrak{g}^*$, then we can check that $\mu(v) = (u \mapsto \omega(v, u)v) \in \mathfrak{g}$. $\mu(v)$ is nilpotent since $\mu(v)^2 = 0$, and the image of μ is a nonzero nilpotent orbit. It is the minimal nilpotent orbit since $\mu(v)$ has rank 1. G acts transitively on $V \setminus \{0\}$, μ is a double cover onto $\mathbb{O}_{min}$. $\square$

Remark 29.6. $\mathbb{O}_{min} = [2, 1, \cdots, 1]$ and $\pi_1^G(\mathbb{O}_{\mathrm{min}}) = \mathbb{Z}/2\mathbb{Z}$, the cover above is the universal cover of $\mathbb{O}_{min}$.

29.2. **Classical and Quantum Formal Darboux Theorems.** The classical Darboux theorem states that every point in a symplectic manifold has a neighborhood that is symplectomorphic to an open subset of the standard symplectic vector space. Here we investigate a formal algebraic analog of this theorem and its extension to quantization.

Lemma 29.7. *Let A be a Poisson algebra and $\mathfrak{m}$ be its maximal ideal with $A/\mathfrak{m} = \mathbb{C}$. Then the Poisson bracket on A induces a skew-symmetric form on $\mathfrak{m}/\mathfrak{m}^2$.*

Proof. Define the skew-symmetric form as:

$$\omega(a + \mathfrak{m}^2, b + \mathfrak{m}^2) = \{a, b\} \mod \mathfrak{m}.$$

$\square$

Now let $A = \mathbb{C}[[x_1, \dots, x_{2n}]]$ so that there is a unique maximal ideal $\mathfrak{m} = (x_1, \dots, x_{2n})$ and $\mathfrak{m}/\mathfrak{m}^2$ is spanned by $x_i + \mathfrak{m}^2$. Suppose that the Poisson bracket on A induces a non-degenerate skew-symmetric form on $\mathfrak{m}/\mathfrak{m}^2$. So $\{x_i, x_j\} = \omega_{ij} \mod \mathfrak{m}$ for some $\omega_{ij} \in \mathbb{C}$, then the matrix (ω_{ij}) is non-degenerate skew symmetric.

Theorem 29.8 (Formal Darboux Theorem)**.** *There exists a formal change of coordinates $y_i = y_i(x_1, \dots, x_{2n}) \in \mathfrak{m}$ such that $\{y_i, y_j\} = \omega_{ij} \in \mathbb{C}$.*

Proof. Use induction, suppose that we have found $x_i^{(n)} \in \mathfrak{m}$ such that $\left\{x_i^{(n)}, x_j^{(n)}\right\} \equiv \omega_{ij} \mod \mathfrak{m}^n$ (i.e. $\left\{x_i^{(n)}, x_j^{(n)}\right\} = \omega_{ij} + R_{ij}^{(n)}$ for some $R_{ij}^{(n)} \in \mathfrak{m}^n$). We want to find $\delta_i^{(n)} \in \mathfrak{m}^{n+1}$ such that $\left\{x_i^{(n)} + \delta_i^{(n)}, x_j^{(n)} + \delta_j^{(n)}\right\} \equiv \omega_{ij} \mod \mathfrak{m}^{n+1}$.

Since

$$\begin{aligned}\left\{x_i^{(n)} + \delta_i^{(n)}, x_j^{(n)} + \delta_j^{(n)}\right\} &= \left\{x_i^{(n)}, x_j^{(n)}\right\} + \left\{\delta_i^{(n)}, x_j^{(n)}\right\} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} + \left\{\delta_i^{(n)}, \delta_j^{(n)}\right\} \\ &\equiv \left\{x_i^{(n)}, x_j^{(n)}\right\} + \left\{\delta_i^{(n)}, x_j^{(n)}\right\} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} \mod \mathfrak{m}^{n+1}\end{aligned}$$

$$\equiv \omega_{ij} + R_{ij}^{(n)} + \left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} \mod \mathfrak{m}^{n+1}.$$

We need to solve the following equation for $\delta_i^{(n)}$:

$$\left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} = -R_{ij}^{(n)} \mod \mathfrak{m}^{n+1}.$$

Note that $\left\{x_i^{(n)}, \delta_j^{(n)}\right\} - \left\{x_j^{(n)}, \delta_i^{(n)}\right\} = \omega_{ik} x_k^{(n)} \frac{\partial \delta_j^{(n)}}{\partial x_k} - \omega_{jk} x_k^{(n)} \frac{\partial \delta_i^{(n)}}{\partial x_k}$, we can solve the above equation by setting

$$\delta_i^{(n)} = -\frac{1}{n+1} \sum_{k,l} \omega^{kl} x_k^{(n)} R_{li}^{(n)}.$$

Set $x_i^{(n+1)} = x_i^{(n)} + \delta_i^{(n)}$, then $y_i = \lim_{n\to\infty} x_i^{(n)}$ satisfies $\{y_i, y_j\} \in \mathbb{C}$. □

Theorem 29.9 (Quantum Formal Darboux Theorem)**.** *Let $\mathcal{A}_\hbar$ be a formal quantization of the Poisson algebra A. Then there are elements $\widehat{y}_i \in \mathcal{A}_\hbar$ such that*

- $\widehat{y}_i \equiv x_i \mod \hbar\mathcal{A}_\hbar$*;*
- $\frac{1}{\hbar}\left[\widehat{y}_i, \widehat{y}_j\right] \in \mathbb{C}$.

Proof. Choose lift of y_i in $\mathcal{A}_\hbar$, denoted by $\widehat{y}_i^{(1)}$. Then $\frac{1}{\hbar}\left[\widehat{y}_i^{(1)}, \widehat{y}_j^{(1)}\right] \equiv \{y_i, y_j\} = \omega_{ij} \mod \hbar$, so we can write $\frac{1}{\hbar}\left[\widehat{y}_i^{(1)}, \widehat{y}_j^{(1)}\right] = \omega_{ij} + \hbar R_{ij}^{(1)}$ for some $R_{ij}^{(1)} \in \mathcal{A}_\hbar$ and $\omega_{ij} \in \mathbb{C}$.

Again, use induction. Suppose that we have found a lift $\widehat{y}_i^{(n)}$ of y_i such that

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \widehat{y}_j^{(n)}\right] = \omega_{ij} + \hbar^n R_{ij}^{(n)}$$

for some $R_{ij}^{(n)} \in \mathcal{A}_\hbar$. We want to find $\delta_i^{(n)} \in \hbar^{n+1}\mathcal{A}_\hbar$ such that

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)} + \delta_i^{(n)}, \widehat{y}_j^{(n)} + \delta_j^{(n)}\right] = \omega_{ij} + \hbar^{n+1} R_{ij}^{(n+1)}.$$

This is equivalent to solving the following equation for $\delta_i^{(n)}$:

$$\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \delta_j^{(n)}\right] - \frac{1}{\hbar}\left[\widehat{y}_j^{(n)}, \delta_i^{(n)}\right] = -\hbar^n R_{ij}^{(n)} \mod \hbar^{n+1}.$$

Note that $\frac{1}{\hbar}\left[\widehat{y}_i^{(n)}, \delta_j^{(n)}\right] - \frac{1}{\hbar}\left[\widehat{y}_j^{(n)}, \delta_i^{(n)}\right] = \omega_{ik}\widehat{y}_k^{(n)} \frac{\partial \delta_j^{(n)}}{\partial \widehat{y}_k} - \omega_{jk}\widehat{y}_k^{(n)} \frac{\partial \delta_i^{(n)}}{\partial \widehat{y}_k}$, we can solve the above equation by setting

$$\delta_i^{(n)} = -\frac{1}{n+1} \sum_{k,l} \omega^{kl} \widehat{y}_k^{(n)} R_{li}^{(n)}.$$

Set $\widehat{y}_i^{(n+1)} = \widehat{y}_i^{(n)} + \delta_i^{(n)}$, then $\widehat{y}_i = \lim_{n\to\infty} \widehat{y}_i^{(n)}$ satisfies $\frac{1}{\hbar}\left[\widehat{y}_i, \widehat{y}_j\right] = \omega_{ij} \in \mathbb{C}$. □

29.3. **Localness of Differential Operators.**

Any differential operator on A can extend to localizations and completions.

- Extension to the Localization $S^{-1}A$. Let A be a commutative algebra over a field k. A k-linear map $L : A \to A$ is a differential operator of order $\leq k$ if for any $a_0, \ldots, a_k \in A$, the iterated commutator vanishes:

$$[\ldots[[L, a_0], a_1], \ldots, a_k] = 0$$

where $[L,a](b) = L(ab) - aL(b)$. We denote the space of such operators as $\mathrm{Diff}^k(A)$.The General Extension FormulaFor any $L \in \mathrm{Diff}^k(A)$, the unique extension $\widetilde{L} \in \mathrm{Diff}^k(S^{-1}A)$ is determined by its action on the reciprocal of an element $s \in S$. The formula uses the iterated adjoint action $\mathrm{ad}_s(L) = [L,s]$:

$$\widetilde{L}\left(\frac{1}{s}\right) = \sum_{j=0}^{k} (-1)^j s^{-(j+1)} \mathrm{ad}_s^j(L)(1)$$

Applying this to a general fraction $\frac{a}{s}$ via the Leibniz-like property for higher-order operators, we get:

$$\widetilde{L}\left(\frac{a}{s}\right) = \sum_{j=0}^{k} (-1)^j s^{-(j+1)} \mathrm{ad}_s^j(L)(a).$$

- Specific Cases for Low OrderOrder 0 (Multiplication by $f \in A$):

$$\widetilde{L}\left(\frac{a}{s}\right) = s^{-1}(f \cdot a) = \frac{fa}{s}$$

Order 1 (Derivations, where $L(1) = 0$):Using the general formula for $j = 0, 1$:

$$\widetilde{L}\left(\frac{a}{s}\right) = s^{-1}L(a) - s^{-2}[L,s](a) = \frac{L(a)}{s} - \frac{L(sa) - sL(a)}{s^2} = \frac{sL(a) - aL(s)}{s^2}$$

Order 2 (e.g., a Laplacian Δ):

$$\widetilde{L}\left(\frac{a}{s}\right) = \frac{\Delta(a)}{s} - \frac{\Delta(sa) - s\Delta(a)}{s^2} + \frac{\Delta(s^2a) - 2s\Delta(sa) + s^2\Delta(a)}{s^3}.$$

- Extension to the Completion $\widehat{A}$. Let $I \subset A$ be an ideal. The completion is the inverse limit $\widehat{A} = \varprojlim A/I^n$.The Continuity PrecisionA differential operator L of order k is strictly continuous with respect to the I-adic topology because it satisfies:

$$L(I^n) \subseteq I^{n-k} \quad \text{for all } n \geq k$$

This filtration ensures that for any Cauchy sequence (a_n) in A, the sequence $(L(a_n))$ is also Cauchy in the I-adic sense. The extension $\widehat{L} : \widehat{A} \to \widehat{A}$ is defined by taking the limit of the action on the truncated sequences:

$$\widehat{L}\left(\lim_{n\to\infty} a_n\right) = \lim_{n\to\infty} L(a_n)$$

Because L is a polynomial-type operator (it "sees" only a finite neighborhood of any point algebraically), this limit is always well-defined and unique.

- The Poisson Bracket CaseA Poisson bracket $\{\cdot,\cdot\}$ is a bilinear map that is a derivation (order 1 operator) in each slot.On $S^{-1}A$: Since $L_f = \{f,\cdot\}$ is a derivation, the extension is forced by the first-order formula derived above. For $a/s, b/t \in S^{-1}A$:

$$\left\{\frac{a}{s}, \frac{b}{t}\right\} = \frac{1}{s^2t^2}\left(st\{a,b\} - at\{s,b\} - bs\{a,t\} + ab\{s,t\}\right)$$

On $\widehat{A}$: Since the bracket is a derivation in each variable, it is continuous in each variable. By the universal property of completions of bilinear maps, it extends uniquely:

$$\{\widehat{a},\widehat{b}\}_{\widehat{A}} = \lim_{n\to\infty} \{a_n, b_n\}_A$$

This ensures that $\widehat{A}$ inherits the Poisson algebra structure from A without any ambiguity.

30. Normality of Nilpotent Orbit Closures

2026-04-24

Normality of Closure of Nilpotent Orbits

Theorem 30.1 (Main theorem)**.** *For a nilpotent orbit $\mathbb{O}$ in a complex semisimple Lie algebra $\mathfrak{g}$, its closure $\overline{\mathbb{O}}$ is normal if and only if $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$.*

Lemma 30.2 (Serre's criterion for normality)**.** *A is a commutative Noetherian ring. Then A is normal if and only if it satisfies the following two conditions:*

(R_1) *For any prime ideal $\mathfrak{p}$ of A with $\mathrm{ht}(\mathfrak{p}) \leq 1$, the localization $A_{\mathfrak{p}}$ is a regular local ring.*
(S_2) *For any prime ideal $\mathfrak{p}$ of A, we have $\operatorname{depth}(A_{\mathfrak{p}}) \geq \min\{2, \mathrm{ht}(\mathfrak{p})\}$.*

Proof of Main Theorem. ($\Rightarrow$) If $\overline{\mathbb{O}}$ is normal, then by Serre's criterion, we get information about the depth inequality, and then we can apply the algebraic Hartogs theorem [Har77, Chapter III, Exercise 3.5] to conclude that $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$.

($\Leftarrow$) If $\mathbb{C}[\mathbb{O}] = \mathbb{C}[\overline{\mathbb{O}}]$, then $\mathbb{C}[\overline{\mathbb{O}}]$ is a domain and integrally closed in its field of fractions, Since $\mathbb{C}[\mathbb{O}]$ is smooth. Hence $\mathbb{C}[\overline{\mathbb{O}}]$ is normal. $\square$

31. Filtered Quantization of the Nilpotent Cone

2026-04-25

[Los22]

Lemma 31.1. *Let A be a filtered algebra, $I \subseteq A$ an ideal, we have a short exact sequence:*

$$0 \to \operatorname{gr} I \to \operatorname{gr} A \to \operatorname{gr}(A/I) \to 0.$$

Proof. I has an induced filtration $I_{\leq n} = I \cap A_{\leq n}$, and $\operatorname{gr} I = \bigoplus_n I_{\leq n}/I_{\leq n-1}$ is a graded ideal of $\operatorname{gr} A$. It is easy to see that the map $\operatorname{gr} A \to \operatorname{gr}(A/I)$ is surjective, and $\operatorname{gr} I \to \operatorname{gr} A$ is injective. For an element $a + A_{\leq i-1} \in \operatorname{gr}_i A$, then

$$\begin{aligned} & a + A_{\leq i-1} \mapsto 0 \in \operatorname{gr}_i(A/I) \\ \Leftrightarrow\quad & \overline{a} \in (A/I)_{\leq i-1} \\ \Leftrightarrow\quad & \exists\, a' \in A_{\leq i-1} \text{ such that } a - a' \in I \\ \Leftrightarrow\quad & a + A_{\leq i-1} \in I_{\leq i}/I_{\leq i-1}. \end{aligned}$$

$\square$

Remark 31.2. If $I = \langle f_1, \cdots, f_n\rangle$, then $\langle \sigma(f_1), \cdots, \sigma(f_n)\rangle \subseteq \operatorname{gr} I$.

Proposition 31.3. *A is a $\mathbb{Z}_{\geq 0}$ graded Noetherian Poisson algebra with $\deg\{,\} = -d < 0$, $\mathcal{A}$ is its filtered quantization, and $\mathcal{I} = \langle \widehat{f}_1, \cdots, \widehat{f}_n\rangle$ ($\widehat{f}_i \in \mathcal{A}_{d_i}$), $f_i = \widehat{F}_i + \mathcal{A}_{\leq d_i - 1}$, with the following conditions:*

- *$f_1, \cdots, f_n$ is a regular sequence in A;*
- *$\left[\widehat{f}_i, \widehat{f}_j\right] = \sum_{i=1}^{r} \widehat{c}_{ij}^{k} \widehat{f}_k$, with $\widehat{c}_{ij}^{k} \in \mathcal{A}_{\leq d_i + d_j - d - d_k}$.*

Then $\operatorname{gr} \mathcal{I} = \langle f_1, \cdots, f_n\rangle$.

Proof. We only need to check that $\operatorname{gr}\mathcal{I} \subseteq \langle f_1, \cdots, f_n\rangle$. Assume on the contrary that there exists a positive integer e and $\widehat{b}_i \in \mathcal{A}_{\leq e-d_i}$ such that $\sigma(\sum_{i=1}^n \widehat{b}_i \widehat{f}_i) \notin \langle f_1, \cdots, f_n\rangle$. Also assume e is the smallest integer with this property.

Note that $\sum_{i=1}^n b_i f_i \in A_e$ must be zero, otherwise $\sigma(\sum_{i=1}^n \widehat{b}_i \widehat{f}_i) = \sum_{i=1}^n b_i f_i \in \langle f_1, \cdots, f_n\rangle$. So by the vanishing of the first Koszul homology, we can find $b_{ij} \in A$ such that $b_i = \sum_{j=1}^n b_{ij} f_j$ and $b_{ij} = -b_{ji}$ for $i \neq j$. By restriction to the degree d_i we can assume that $b_{ij} \in A_{\leq e-d_i-d_j}$, take lifts $\widehat{b}_{ij} \in \mathcal{A}_{\leq e-d_i-d_j}$, such that $\widehat{b}_{ij} = -\widehat{b}_{ji}$. Then $\widehat{b}'_i := \widehat{b}_i - \sum_{j=1}^n \widehat{b}_{ij} \widehat{f}_j \in \mathcal{A}_{<e-d_i}$.

$$\begin{aligned}
\sum_{i=1}^n \widehat{b}_i \widehat{f}_i &= \sum_{i=1}^n \widehat{b}'_i \widehat{f}_i + \sum_{i,j=1}^n \widehat{b}_{ij} \widehat{f}_j \widehat{f}_i \\
&= \sum_{i=1}^n \widehat{b}'_i \widehat{f}_i + \sum_{j<k} \widehat{b}_{jk} [\widehat{f}_j, \widehat{f}_k] \\
&= \sum_{i=1}^n \widehat{b}'_i \widehat{f}_i + \sum_{j<k} \widehat{b}_{jk} \sum_{i=1}^n \widehat{c}^i_{jk} \widehat{f}_i \\
&= \sum_{i=1}^n \left(\widehat{b}'_i + \sum_{j<k} \widehat{b}_{jk} \widehat{c}^i_{jk} \right) \widehat{f}_i.
\end{aligned}$$

But $\widehat{b}'_i + \sum_{j<k} \widehat{b}_{jk} \widehat{c}^i_{jk} \in \mathcal{A}_{<e-d_i}$, which is a contradiction to the minimality of e. $\square$

A discovery: Harish-Chandra's isomorphism can be viewed as a quantized version of the Chevalley restriction theorem.

Theorem 31.4 (Chevalley restriction theorem)**.** *Let $\mathfrak{g}$ be a complex reductive Lie algebra, $\mathfrak{h}$ be a Cartan subalgebra, W be the Weyl group. Then the restriction map $\mathbb{C}[\mathfrak{g}]^G \to \mathbb{C}[\mathfrak{h}]^W$ is an isomorphism.*

Theorem 31.5 (Harish-Chandra isomorphism)**.** *Let $\mathfrak{g}$ be a complex reductive Lie algebra, $\mathfrak{h}$ be a Cartan subalgebra, W be the Weyl group. Then there is an isomorphism $Z(\mathfrak{g}) = \mathcal{U}(\mathfrak{g})^G \to \mathbb{C}[\mathfrak{h}]^W$.*

Corollary 31.6. *$\mathfrak{h}/W$ parametrizes both the filtered Poisson deformations and the filtered quantizations of $\mathcal{N}$.*

Some phenomena of separation of variables:

- $\mathbb{C}[\mathfrak{g}] = \mathbb{C}[\mathfrak{g}]^G \otimes_{\mathbb{C}} \mathcal{H}$;
- $\mathbb{C}[\mathfrak{h}] = \mathbb{C}[\mathfrak{h}]^W \otimes_{\mathbb{C}} \mathcal{H}$;
- $\mathbb{C}[V] = \mathbb{C}[V]^{\mathrm{SO}(V)} \otimes_{\mathbb{C}} \mathcal{H}(V) = \mathbb{C}[r^2] \otimes_{\mathbb{C}} \mathcal{H}(V)$. (Spherical harmonics)

Proof of the first separation of variables. There is an isomorphism $\mathbb{C}[\mathfrak{g}] = \mathbb{C}[\mathfrak{g}]^G_+ \mathbb{C}[\mathfrak{g}] \oplus \mathcal{H}$.

Choose a homogeneous vector space basis $\overline{b}_i$ of $\mathbb{C}[\mathfrak{g}]/(\mathbb{C}[\mathfrak{g}]^G_+\mathbb{C}[\mathfrak{g}])$, and let b_i be a homogeneous lift of $\overline{b}_i$ in $\mathcal{H}$. Then b_i is a vector space basis of $\mathcal{H}$. By graded Nakayama's lemma, b_i generates $\mathbb{C}[\mathfrak{g}]$ as a $\mathbb{C}[\mathfrak{g}]^G$-module, we claim that b_i is a free basis of $\mathbb{C}[\mathfrak{g}]$ as a $\mathbb{C}[\mathfrak{g}]^G$-module, so that the following is an isomorphism

$$\mathbb{C}[\mathfrak{g}]^G \otimes_{\mathbb{C}} \mathcal{H} \to \mathbb{C}[\mathfrak{g}], \quad f \otimes b \mapsto fb.$$

If not, then there exists $r_i \in \mathbb{C}[\mathfrak{g}]^G$ such that $\sum_i r_i b_i = 0$, and not all r_i are zero. Projection to $\mathcal{H}$ we have $\sum_i \overline{r}_i \overline{b}_i = 0$, since $\overline{b}_i$ is a vector space basis, so $\overline{r}_i = 0$ for all i which means $r_i \in \mathbb{C}[\mathfrak{g}]^G_+\mathbb{C}[\mathfrak{g}] = \langle f_1, \cdots, f_n\rangle$, assume that $r_i = \sum_j r_{ij} f_j$ then

$$0 = \sum_i r_i b_i = \sum_{i,j} r_{ij} f_j b_i = \sum_j f_j \underbrace{\left(\sum_i r_{ij} b_i\right)}_{a_j}.$$

Since f_i form a regular sequence, the first Koszul homology vanishes, which means that there exists s_{jk} such that $a_j = \sum_k s_{jk} f_k$ and $s_{jk} = -s_{kj} \in \mathbb{C}[\mathfrak{g}]$. So we have

$$\sum_k s_{jk} f_k = a_j = \sum_i r_{ij} b_i.$$

This implies that $r_{ij} \in \langle f_1, \cdots, f_n\rangle$, suppose $r_{ij} = \sum_k r_{ijk} f_k$, then

$$0 = \sum_{i,j} r_{ij} f_j b_i = \sum_{i,j,k} r_{ijk} f_k f_j b_i = \sum_{j,k} f_j f_k \underbrace{\left(\sum_i r_{ijk} b_i\right)}_{a_{jk}}.$$

Again by the vanishing of the second Koszul homology, there exists s_{jkl} such that $a_{jk} = \sum_l s_{jkl} f_l$ and $s_{jkl} = -s_{jlk}$, so we have

$$\sum_l s_{jkl} f_l = a_{jk} = \sum_i r_{ijk} b_i.$$

This implies that $r_{ijk} \in \langle f_1, \cdots, f_n\rangle$, and we can continue this process to conclude that $r_i \in \langle f_1, \cdots, f_n\rangle^m$ for all m, which implies that $r_i = 0$. ☐

32. Divisors and Picard Groups

2026-04-26

Some relations between Picard groups, Cartier divisors, and Weil divisors.

Pic is the group of isomorphism classes of line bundles. Cl is the group of Weil divisors modulo linear equivalence. CaCl is the group of Cartier divisors modulo linear equivalence.

- If X is integral, then $\mathrm{Pic}(X) \cong \mathrm{CaCl}(X)$.
- If X is integral and normal, then $\mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X)$.
- If X is integral and locally factorial/Smooth, then $\mathrm{Pic}(X) \cong \mathrm{Cl}(X)$.

If X is integral and normal, by Serre's criterion for normality, the singular locus of X has codimension at least 2, hence $\mathrm{Cl}(X) \cong \mathrm{Cl}(X^{reg}) \cong \mathrm{Pic}(X^{reg})$.

33. Symplectic Leaves of Nilpotent Orbits

2026-04-27

[Los22]

We prove that for a nilpotent orbit $\mathbb{O}$ in a complex semisimple Lie algebra $\mathfrak{g}$,

$$\overline{\mathbb{O}}^{reg} = \mathbb{O}.$$

Lemma 33.1. *$\overline{\mathbb{O}}^{reg}$ is symplectic.*

Proof. Since $\overline{\mathbb{O}}$ is Poisson, it has a Poisson bivector field $\pi \in \Gamma(\overline{\mathbb{O}}, \wedge^2 T_{\overline{\mathbb{O}}})$, and π is non-degenerate on $\mathbb{O}$, and we know that the zero locus of a section of a line bundle on the smooth variety $\overline{\mathbb{O}}^{reg}$ has codimension at most 1, but $\operatorname{codim}_{\overline{\mathbb{O}}^{reg}}(\{\pi = 0\}) \geq 2$, so π is non-vanishing on $\overline{\mathbb{O}}^{reg}$. $\square$

Let X be a Poisson variety, and $X_1 = X \backslash X^{reg}$, $X_{k+1} = X_k \backslash X_k^{reg}$. We say that X has finitely many symplectic leaves if each X_k^{reg} is symplectic. For a symplectic leaf, we mean an irreducible (connected) component of X_k^{reg} for some k.

Lemma 33.2. *Symplectic leaves are in one-to-one correspondence with prime Poisson ideals of* $\mathbb{C}[X]$.

Lemma 33.3. $\overline{\mathbb{O}}$ *has finitely many symplectic leaves, and the symplectic leaves are exactly the nilpotent orbits contained in* $\overline{\mathbb{O}}$.

Proof. The second statement is due to the fact that Poisson bracket is generated by Lie bracket. $\square$

Lemma 33.4. *A smooth symplectic variety has no proper Poisson subvarieties, hence it is a symplectic leaf.*

Proof. Suppose X is a smooth symplectic variety, and let $x \in X$ be a point.

$$T_x X \xrightarrow{\pi_x} T_x^* X$$

is an isomorphism.

Under this isomorphism H_f corresponds to $\mathrm{d}f$, so we can see that $\operatorname{span}\{H_f \mid f \in \mathbb{C}[X]\} = T_x X$, but Poisson subvariety must tangent to all Hamiltonian vector fields, so it must be the whole X. $\square$

Proof of $\overline{\mathbb{O}}^{reg} = \mathbb{O}$. Since $\overline{\mathbb{O}}$ has finitely many symplectic leaves, $\overline{\mathbb{O}}^{reg}$ is a symplectic leaf, hence it is a nilpotent orbit. Since $\mathbb{O} \subseteq \overline{\mathbb{O}}^{reg}$, we have $\overline{\mathbb{O}}^{reg} = \mathbb{O}$. $\square$

Remark 33.5. The argument above works for any coadjoint orbits in a complex Lie algebra, not necessarily semisimple.

Remark 33.6. For a nilpotent orbit cover $\widetilde{\mathbb{O}}$, we have a finite surjective morphism

$$\pi : \operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}]) \to \overline{\mathbb{O}},$$

$\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ has finitely many symplectic leaves, and we also know that $\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])^{reg}$ is symplectic, it has no proper Poisson subvarieties, hence it is a symplectic leaf. By in this situation, symplectic leaves are not in one-to-one correspondence with G-orbits any more, so $\widetilde{\mathbb{O}}$ may properly contained in $\operatorname{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])^{reg}$.

Actually in the case of exceptional Lie algebras, there exist some nilpotent orbits $\mathbb{O}$ such that $\operatorname{codim}_{\overline{\mathbb{O}}}(\overline{\mathbb{O}} \backslash \mathbb{O}) = 2$ but $\operatorname{codim}_X(X \backslash X^{reg}) \geq 4$, where $X = \operatorname{Spec}(\mathbb{C}[\mathbb{O}])$.

34. Richardson Orbits and Symplectic Resolutions

2026-04-27

34.1. **A fundamental fact I often forget.** G be a complex connected reductive algebraic, $\mathcal{B}$ be the flag variety of G, then

$$T\mathcal{B} = \{(b,\xi)\,|\, b \in \mathcal{B}, \quad \xi \in \mathfrak{g}/b\}.$$

For any $X \in \mathfrak{g}$, the vector field $X^\sharp$ on $\mathcal{B}$ generated by X-action is given by

$$X^\sharp(b) = (b, X + b).$$

We also have the cotangent bundle

$$T^*\mathcal{B} = \{(b,\xi)\,|\, b \in \mathcal{B}, \quad \xi \in b^\perp \subseteq \mathfrak{g}^*\}.$$

For a symmetric subgroup K of G, we have:

$$T^*_K\mathcal{B} = \{(b,\xi)\,|\, b \in \mathcal{B}, \quad \xi \in b^\perp \cap \mathfrak{k}^\perp\}.$$

We have the following Cartesian diagram:

$$\begin{tikzcd}
T^*_K\mathcal{B} \arrow[r, hook] \arrow[d] & T^*\mathcal{B} \arrow[d] \\
\mathfrak{k}^\perp \arrow[r, hook] & \mathfrak{g}^*
\end{tikzcd}$$

34.2. **Richardson orbits.** For a parabolic $P \subseteq G$, $T^*(G/P) = T^*\mathcal{P}$ has a unique open G-orbit.

Since there is a bijection between P-orbits in $\mathfrak{u}$ and G-orbits in $T^*\mathcal{P}$, the unique open G-orbit in $T^*\mathcal{P}$ corresponds to the unique open P-orbit in $\mathfrak{u}$.

We also have the moment map $\mu : T^*\mathcal{P} \to \mathfrak{g}^*$, the image of μ is the closure of a nilpotent orbit $\mathbb{O}_{\mathcal{P}}$, which is called the Richardson orbit associated to $\mathcal{P}$.

We have the following Cartesian diagram:

$$\begin{tikzcd}
\widetilde{\mathbb{O}}_{\mathcal{P}} \arrow[r, hook] \arrow[d, "\mu"] & T^*\mathcal{P} \arrow[d, "\mu"] \\
\mathbb{O}_{\mathcal{P}} \arrow[r, hook] & \overline{\mathbb{O}}_{\mathcal{P}}
\end{tikzcd}$$

where $\widetilde{\mathbb{O}}_{\mathcal{P}}$ is the unique open G-orbit in $T^*\mathcal{P}$, it is also a finite cover of $\mathbb{O}_{\mathcal{P}}$.

When the orbit $\mathbb{O}_{\mathcal{P}}$ is even, the moment map μ is just the Jacobson-Morozov resolution of $\overline{\mathbb{O}}_{\mathcal{P}}$, and $\widetilde{\mathbb{O}}_{\mathcal{P}} = \mathbb{O}_{\mathcal{P}}$. So all the even orbits are Richardson orbits, and they admit symplectic resolutions.

35. Equivariant Structures on the Structure Sheaf

2026-04-29

Let Γ be a discrete group acting on a complex algebraic variety X, we want to classify the Γ-equivariant structures on the structure sheaf $\mathcal{O}_X$. (Note that in the case of discrete group actions, the Γ-equivariant structure on a sheaf $\mathcal{F}$ is just a collection of isomorphisms $\phi_\gamma : \mathcal{F} \to \gamma^*\mathcal{F}$ for each $\gamma \in \Gamma$, such that the cocycle condition holds.[Ach21])

Lemma 35.1. *There is a natural Γ-equivariant structure on $\mathcal{O}_X$ given by the translation of functions, i.e. for $g \in \Gamma$, the isomorphism $T_\gamma : \mathcal{O}_X \to \gamma^*\mathcal{O}_X$ is given by $T_\gamma(f) = f(\gamma^{-1}(\cdot))$.*

Theorem 35.2. *The isomorphism classes of Γ-equivariant structures on $\mathcal{O}_X$ are in one-to-one correspondence with $H^1(\Gamma, \mathcal{O}_X^*(X))$.*

Proof. Let $\phi_\gamma : \mathcal{O}_X \to \gamma^*\mathcal{O}_X$ be a Γ-equivariant structure on $\mathcal{O}_X$, compose with the inverse translation isomorphism $T_\gamma^{-1} : \gamma^*\mathcal{O}_X \to \mathcal{O}_X$, we get an automorphism $\psi_\gamma = T_\gamma^{-1} \circ \phi_\gamma$ of $\mathcal{O}_X$, which is given by multiplication by some invertible function $f_\gamma \in \mathcal{O}_X^*(X)$. At this time, ϕ is given by the explicit formula

$$\begin{array}{cccc} \phi_\gamma : & \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_X(\gamma \cdot U) \\ & f & \longmapsto & f_\gamma(\gamma^{-1}(\cdot)) f(\gamma^{-1}(\cdot)) \end{array}$$

The compatibility condition explicitly says that the following two morphisms are equal:

$$\begin{tikzcd} \mathcal{O}_X(U) \arrow[r, "\phi_\gamma"] & \mathcal{O}_X(\gamma \cdot U) \arrow[r, "\phi_\delta"] & \mathcal{O}_X(\delta\gamma \cdot U) \\ f \arrow[r, mapsto] & f_\gamma(\gamma^{-1}(\cdot)) f(\gamma^{-1}(\cdot)) \arrow[r, mapsto] & f_\delta(\delta^{-1}(\cdot)) f_\gamma(\gamma^{-1}\delta^{-1}(\cdot)) f(\gamma^{-1}\delta^{-1}(\cdot)) \end{tikzcd}$$

$$\begin{tikzcd} \mathcal{O}_X(U) \arrow[r, "\phi_{\delta\gamma}"] & \mathcal{O}_X(\delta\gamma \cdot U) \\ f \arrow[r, mapsto] & f_{\delta\gamma}((\delta\gamma)^{-1}(\cdot)) f((\delta\gamma)^{-1}(\cdot)) \end{tikzcd}$$

This means that $f_{\delta\gamma} = f_\delta \cdot (\delta \cdot f_\gamma)$, so f_γ is a 1-cocycle of Γ with coefficients in $\mathcal{O}_X^*(X)$.

Conversely, given a 1-cocycle f_γ, we can define a Γ-equivariant structure on $\mathcal{O}_X$ by the explicit formula above, and the compatibility condition is automatically satisfied.

Finally, two Γ-equivariant structures ϕ and ϕ' are isomorphic if and only if there exists an automorphism $\psi : \mathcal{O}_X \to \mathcal{O}_X$ such that the following diagram commutes for all $\gamma \in \Gamma$:

$$\begin{tikzcd} \mathcal{O}_X \arrow[r, "\phi_\gamma"] \arrow[d, "\psi"] & \gamma^*\mathcal{O}_X \arrow[d, "\gamma^*\psi"] \\ \mathcal{O}_X \arrow[r, "\phi'_\gamma"] & \gamma^*\mathcal{O}_X \end{tikzcd}$$

Since the automorphism ψ is given by multiplication by some invertible function $h \in \mathcal{O}_X^*(X)$, the above diagram commutes if and only if $\frac{f'_\gamma}{f_\gamma} = \frac{h(\gamma^{-1}(\cdot))}{h(\cdot)}$, which means that f_γ and f'_γ are cohomologous. □

Corollary 35.3. *If $\mathcal{O}_X^*(X) = \mathbb{C}^*$, then all the Γ-equivariant structures on $\mathcal{O}_X$ is given by characters in $\mathfrak{X}(\Gamma)$: $\mathbb{C}_\chi \mapsto \mathbb{C}_\chi \otimes \mathcal{O}_X$*

Proof.

$$\mathfrak{X}(\Gamma) = \operatorname{Hom}(\Gamma, \mathbb{C}^*) \cong H^1(\Gamma, \mathcal{O}_X^*(X)).$$

□

36. Principal Nilpotent Elements

2026-04-30

Let $\mathfrak{g}$ be a complex reductive Lie algebra, fix a Borel $\mathfrak{b}$ and a Cartan $\mathfrak{h} \subseteq \mathfrak{b}$, we have the root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$, where Φ is the set of roots. $\Pi \subseteq \Phi^+$ be the set of simple roots, for each $\alpha \in \Pi$, we can choose a nonzero root vector $X_\alpha \in \mathfrak{g}_\alpha$, then

$$X = \sum_{\alpha\in\Pi} X_\alpha$$

lies in the principal nilpotent orbits $\mathbb{O}_{prin}$.

Let $h = \sum_{\alpha\in\Phi^+} \check{\alpha}$

Lemma 36.1. $[h, X] = 2X$

Proof. For each $\alpha \in \Pi$, we have $\alpha(h) = 2$, since $s_\alpha(h) = h - 2\check{\alpha}$. □

Suppose that $h = \sum_{\alpha\in\Pi} r_\alpha\check{\alpha}$. For each $\alpha \in \Pi$ we can find $Y_\alpha \in \mathfrak{g}_{-\alpha}$ such that $[X_\alpha, Y_\alpha] = r_\alpha\check{\alpha}$. Set

$$Y = \sum_{\alpha\in\Pi} Y_\alpha,$$

then we have $[h, Y] = -2Y$ and $[X, Y] = h$, so $\{X, h, Y\}$ is an $\mathfrak{sl}_2$-triple.

We have a weight decomposition $\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i$, where $\mathfrak{g}_i = \{x \in \mathfrak{g} \mid [h, x] = ix\}$.

Lemma 36.2. $\mathfrak{g}_0 = \mathfrak{h}$, $\mathfrak{g}_{\geq 2} = \mathfrak{n}$.

Suppose $\mathbb{O} = G \cdot X$. Then the Jacobson-Morozov resolution of $\overline{\mathbb{O}}$ is just the Springer resolution $T^*\mathcal{B} \to \overline{\mathbb{O}}_{prin} = \mathcal{N}$, so $\mathbb{O}$ is the principal nilpotent orbit.

37. Conical Symplectic Singularities

2026-05-15

We summarize some facts about conical symplectic singularities and their partial Poisson resolutions, with emphasis on the case of nilpotent covers of complex semisimple Lie algebras. The reference is [Los22].

All the things are on the classical side, we will discuss the quantum side later.

37.1. **Conical symplectic singularity.**

Definition 37.1. For normal irreducible Poisson variety X, we say that X has symplectic singularity if the following conditions are satisfied:

- The Poisson structure on X^{reg} is nondegenerate denoted by ω^{reg};
- There is a resolution of singularity $\pi : Y \to X$ such that $\pi^*(\omega^{reg}) \in \Gamma(\pi^{-1}(X^{reg}); \Omega^2_Y)$ extends to a 2-form $\widetilde{\omega}$ in $\Gamma(Y; \Omega^2_Y)$. ($\widetilde{\omega}$ is a closed 2-form, but possibly degenerate)

Definition 37.2. A conical symplectic singularity is a normal irreducible Poisson variety with symplectic singularity which is also a conic variety.

Proposition 37.3. *For a nilpotent orbit $\mathbb{O}$, $\mathrm{Spec}(\mathbb{C}[\mathbb{O}])$ is a conic symplectic singularity.*

Proof. X^{reg} is symplectic since the boundary has codimension at least 2. Use the Jacobson-Morozov resolution, we can construct the extension of symplectic form explicitly. □

Proposition 37.4. *For a nilpotent cover $\widetilde{\mathbb{O}}$, $\mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ is a conic symplectic singularity.*

Proof. This follows from the previous proposition and Namikawa's results on symplectic singularities. □

37.2. **Partial Poisson resolution, $\mathbb{Q}$-factorial terminalization.**

Definition 37.5. Let X be a normal Poisson variety. A morphism $\pi : Y \to X$ is called partial Poisson resolution if:

- It's proper and birational;
- It's Poisson.

Proposition 37.6. *If X has symplectic singularity then so does Y.*

Proposition 37.7. *Let X be a conical symplectic singularity, $\pi : Y \to X$ is a partial Poisson resolution with* $\operatorname{codim}_Y(Y^{sing}) \geq 4$. *Then $\mathbb{C}[Y^{reg}] \simeq \mathbb{C}[Y] \simeq \mathbb{C}[X]$ and $H^1(Y^{reg}; \mathcal{O}_Y) = H^2(Y^{reg}; \mathcal{O}_Y) = 0$.*

Proof. The first assertion is due to symplectic singularities have rational singularities, and Hartogs theorem.

The second assertion due to the short exact sequence:

$$R\Gamma_{Y^{sing}} \to \mathrm{id} \to R\iota_*\iota^* \xrightarrow{+1},$$

where $\iota : Y^{reg} \to Y$. And the fact that symplectic singularity is Cohen-Macaulay. $\square$

Theorem 37.8. *Let X be a conical symplectic singularity. For a partial Poisson resolution $\pi : Y \to X$ the following are equivalent:*

- *π is maximal Poisson resolution;*
- *Y is $\mathbb{Q}$-factorial and terminal.*

In this case, Y is called a $\mathbb{Q}$-factorial terminalization of X.

For a nilpotent cover $\widetilde{\mathbb{O}}$, suppose that there exists a nilpotent cover $\widetilde{\mathbb{O}}_L$ of a Levi subalgebra such that $\mathrm{Bind}_L^G(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$. Set $X = \mathrm{Spec}(\mathbb{C}[\mathbb{O}])$, $X_L = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}_L])$, and $Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \mid \alpha|_{\mathfrak{l}} = \mu(x)\}$.

Proposition 37.9. *The moment map $\mu : Y \to \mathfrak{g}^*$ factor through $X \to \mathfrak{g}^*$, and $\mu : Y \to X$ is a partial Poisson resolution.*

Proposition 37.10. *Y is $\mathbb{Q}$-factorial and terminal, if and only if X_L is.*

Theorem 37.11. *X_L is $\mathbb{Q}$-factorial and terminal if and only if $\widetilde{\mathbb{O}}_L$ is birationally rigid.*

37.3. **Poisson deformation.**

Definition 37.12. Let X be a conical symplectic singularity and B is a finitely generated positively graded algebra. A graded Poisson deformation of X over $\mathrm{Spec}(B)$ is a Poisson scheme $X_{\mathrm{Spec}(B)}$ over $\mathrm{Spec}(B)$ and a Poisson isomorphism $X_0 \xrightarrow{\sim} X$ such that:

- $X_{\mathrm{Spec}(B)}$ is flat over $\mathrm{Spec}(B)$;
- $X_{\mathrm{Spec}(B)}$ is graded ($\exists\ \ \mathbb{C}^\times \circlearrowleft X_{\mathrm{Spec}(B)}$), such that $\{,\}$ has degree $-d$ (which is the same as X).

Remark 37.13. B lies in the Poisson center of $X_{\mathrm{Spec}(B)}$, but I don't know if it is contained in the definition.

In the case of nilpotent cover, we can construct a graded Poisson deformation of Y:

$$Y_{\mathfrak{z}} := G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \mid \alpha|_{\mathfrak{l}} - \mu(x) \in \mathfrak{z}\}.$$

Here $\mathfrak{z} = (\mathfrak{l}/[\mathfrak{l}, \mathfrak{l}])^*$

Proposition 37.14. *For any $\lambda \in \mathfrak{z}$, the image of $\mu : Y_\lambda \to \mathfrak{g}^*$ is the closure of a coadjoint orbit $\mathbb{O}_\lambda$, and*

$$\mu : Y_\lambda \to \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}_\lambda}])$$

is a partial Poisson resolution.

Theorem 37.15 (Namikawa: Universal Poisson Deformation)**.** *Set $\mathfrak{h}_X := H^2(Y^{reg}; \mathbb{C})$ for a $\mathbb{Q}$-factorial terminalization of X, then there exist a reflection group $W_X \subseteq \mathrm{GL}(\mathfrak{h}_X)$ and a graded Poisson deformation $X_{\mathfrak{h}_X/W_X}$ with the following universal property:*

For any graded Poisson deformation $X_{\mathrm{Spec}(B)}$ of X there exist a unique graded algebra homomorphism $\mathbb{C}[\mathfrak{h}_X]^{W_X} \to B$ such that there is an isomorphism of graded Poisson deformations

$$\mathrm{Spec}(B) \times_{\mathfrak{h}_X/W_X} X_{\mathfrak{h}_X/W_X} \xrightarrow{\sim} X_{\mathrm{Spec}(B)}.$$

$$\begin{tikzcd}
\mathrm{Spec}(B) \times_{\mathfrak{h}_X/W_X} X_{\mathfrak{h}_X/W_X} \arrow[r] \arrow[d] & X_{\mathfrak{h}_X/W_X} \arrow[d] \\
\mathrm{Spec}(B) \arrow[r, "\exists!"] & \mathfrak{h}_X/W_X
\end{tikzcd}$$

In the case of nilpotent cover $X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$, take birationally rigid nilpotent cover $\widetilde{\mathbb{O}}_L$ such that $\mathrm{Bind}_L^G(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$, we have a Poisson deformation $Y_{\mathfrak{z}}$ of $Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \mid \alpha|_{\mathfrak{l}} = \mu(x)\}$ then $X_{\mathfrak{z}} = \mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])$ is a graded deformation of X, by Namikawa's theorem, there is a unique morphism $\mathfrak{z} \to \mathfrak{h}_X/W_X$

the universal Poisson deformation of X can be constructed as $\mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])/W_X = X_{\mathfrak{z}}/W_X$.

$$\begin{tikzcd}
Y \arrow[r, hook] \arrow[dd] & Y_{\mathfrak{z}} \arrow[d] \\
& X_{\mathfrak{z}} \arrow[d] \\
X \arrow[r, hook] & X_{\mathfrak{z}}/W_X
\end{tikzcd}$$

37.4. **Summarize.**

TABLE 1. Conical Symplectic Singularities: General Theory vs. Nilpotent Covers

	General Theory	**Nilpotent Cover Example**
Singularity	Normal irreducible Poisson variety X with symplectic singularities	$X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$ for a nilpotent cover $\widetilde{\mathbb{O}}$.
Partial Resolution	Proper birational Poisson morphism $\pi : Y \to X$.	Generalized Springer map $\mu : Y \to X$ with $Y = G \times^P (X_L \times \mathfrak{p}^\perp)$.
Terminalization	$\mathbb{Q}$-factorial terminalization $\iff$ maximal partial Poisson resolution.	Y is $\mathbb{Q}$-factorial and terminal $\iff$ $\widetilde{\mathbb{O}}_L$ is birationally rigid.
Graded Deformation	Poisson scheme $X_{\mathrm{Spec}(B)}$ flat over a graded algebra B.	Deformation $Y_{\mathfrak{z}}$ constructed via the center $\mathfrak{z} = (\mathfrak{l}/[\mathfrak{l}, \mathfrak{l}])^*$.
Universal Deformation	Base space $\mathfrak{h}_X/W_X$ where $\mathfrak{h}_X := H^2(Y^{reg}; \mathbb{C})$.	Universal deformation constructed as $X_{\mathfrak{z}}/W_X = \mathrm{Spec}(\mathbb{C}[Y_{\mathfrak{z}}])/W_X$.

38. Zariski's Main Theorem

2026-05-16

Theorem 38.1 (Zariski's Main Theorem)**.** *Let $f : X \to Y$ be a quasi-finite, separated morphism of schemes, where Y is quasi-compact and quasi-separated. Then there exists a factorization:*

$$X \xrightarrow{j} X' \xrightarrow{g} Y$$

where j is an open immersion and g is a finite morphism.

Proof. $f_*\mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras. Let $\mathcal{A}$ be the integral closure of $\mathcal{O}_X$ in $f_*\mathcal{O}_Y$, then take $X' = \underline{\mathrm{Spec}}_Y(\mathcal{A})$. □

The intuition is that quasi-finite morphisms have finite fibers, but the fibers can drop abruptly, the theorem tells us that we can appropriately extend the scheme Y to avoid this bad phenomenon.

Example 38.2. $\mathbb{C}^\times \to \mathbb{C}$ can extend to $\mathbb{C}^\times \hookrightarrow \mathbb{C} \xrightarrow{\sim} \mathbb{C}$.

Example 38.3. For nilpotent cover, the morphism $\widetilde{\mathbb{O}} \to \overline{\mathbb{O}}$ can extend to

$$\widetilde{\mathbb{O}} \hookrightarrow \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}]) \to \overline{\mathbb{O}}.$$

39. Twisted Differential Operators

2026-05-17

X be a smooth variety, $\mathcal{L}$ be a line bundle on X. Choose a local trivialization for $\mathcal{L}$ by affine open subsets:

$$X = \bigcup_i X_i, \quad \phi_i : \mathcal{L}|_{X_i} \xrightarrow{\sim} \mathcal{O}_{X_i}.$$

Suppose that $\phi_j \circ \phi_i^{-1} : \mathcal{O}_{X_{ij}} \xrightarrow{\sim} \mathcal{O}_{X_{ij}}$ is given by $f_{ji} \in \mathcal{O}^{\times}(X_{ij})$.

We can define a sheaf of twisted differential operators $\mathcal{D}_{\mathcal{L}}$ via the local trivialization:

$$\psi_i : \mathcal{D}_{\mathcal{L}} \xrightarrow{\sim} \mathcal{D}_X|_{X_i},$$

such that $\psi_j \circ \psi_i^{-1} : \mathcal{D}_{X_{ij}} \xrightarrow{\sim} \mathcal{D}_{X_{ij}}$ is given by $f \mapsto f$ and $\xi \mapsto \xi - \frac{\xi(f_{ji})}{f_{ji}}$.

Theorem 39.1. *$\mathcal{D}_{\mathcal{L}}$ acts on $\mathcal{L}$.*

Proof. For each local trivialization X_i and $X \in \mathcal{D}_{\mathcal{L}}(X_i)$, $s \in \mathcal{L}(X_i)$, define

$$X \cdot s = \phi_i^{-1}(\psi_i(X) \cdot \phi_i(s)).$$

We check it is well defined on the overlap of X_i and X_j, i.e.

$$\phi_j \circ \phi_i^{-1}(\psi_i(X) \cdot \phi_i(s)) = \psi_j(X) \cdot \phi_j(s).$$

If $\psi_i(X) = f \in \mathcal{O}(X_{ij})$, the verification is easy.

If $\psi(X) = \xi \in \mathrm{Vect}(X_{ij})$ is a vector field, then left hand side is

$$f_{ji}\xi \cdot \phi_i(s).$$

Right hand side is

$$(\xi - \frac{\xi(f_{ji})}{f_{ji}}) \cdot (f_{ji} \cdot \phi_i(s)) = \xi(f_{ji})\phi_i(s) + f_{ji}\xi(\phi_i(s)) - \xi(f_{ji})\phi_i(s) = f_{ji}\xi \cdot \phi_i(s).$$

$\square$

40. Quantization of Nonaffine Varieties

2026-05-18

40.1. **Deformation and Quantization of nonaffine schemes.** Let X be a graded normal quasi-projective Poisson variety, the Poisson bracket has degree $-d$. According to Sumihiro's Theorem, any normal variety X with a $\mathbb{G}_m$-action can be covered by $\mathbb{G}_m$-invariant affine open subvarieties U_i.

Definition 40.1. A filtered quantization of $\mathcal{O}_X$ is a pair $(\mathcal{D}, \theta)$ where

- $\mathcal{D}$ is a filtered sheaf of algebra $\mathcal{D}$ in the conical topology of X such that $[\mathcal{D}_{\leq i}, \mathcal{D}_{\leq j}] \subseteq \mathcal{D}_{\leq i+j-d}$;
- $\theta : \mathrm{gr}(\mathcal{D}) \xrightarrow{\sim} \mathcal{O}_X$ is a graded Poisson isomorphism.

Remark 40.2 (A very important situation)**.** When $X = \underline{\mathrm{Spec}}_Y(\mathcal{A})$ for a variety Y and a quasi-coherent graded ring of Poisson algebra $\mathcal{A}$ on Y, we can focus on the variety Y: the filtered quantization of $\mathcal{O}_X$ on X is the same as filtered quantization of $\mathcal{A}$ on Y.

Similarly, we can define filtered Poisson deformation of X.

40.2. **A very important example of Hamiltonian reduction.** Y be an algebraic variety, H is an algebraic group. $\pi : \widetilde{Y} \to Y$ is a principal H-bundle. Then we have the following reduction:

Theorem 40.3. *$T^*(\widetilde{Y}) /\!/\!/_0\, H \xrightarrow{\sim} T^*Y$.*

Proof. The moment map $\mu : T^*\widetilde{Y} \to \mathfrak{h}^*$ is given by $\langle \mu(\alpha), X\rangle = X^\sharp_{\widetilde{Y}}(\alpha)$, so $\mu^{-1}(0)$ consists of the cotangent vector fields which vanish on the vertical tangent spaces:

$$\mu^{-1}(0) = \widetilde{Y} \times_Y T^*Y.$$

$\square$

Theorem 40.4. $\mathcal{D}_{\widetilde{Y}} ///_\lambda H \xrightarrow{\sim} \mathcal{D}^\lambda_Y$.

Here $\mathcal{D}^\lambda_Y$ is the twisted differential operators on $\mathcal{L}^\lambda = \widetilde{Y} \times^H \mathbb{C}_\lambda$.

These theorems verify the quantization commutes with reduction principle.

40.3. Quantization of nilpotent covers.

Suppose $\mathrm{Bind}_L^G(\widetilde{\mathbb{O}}_L) = \widetilde{\mathbb{O}}$, where $\widetilde{\mathbb{O}}_L$ is birationally rigid.

$Y = G \times^P \{(\alpha, x) \in \mathfrak{n}^\perp \times X_L \,|\, \alpha|_{\mathfrak{l}} = \mu(x)\}$ is a $\mathbb{Q}$-factorial terminalization of $X = \mathrm{Spec}(\mathbb{C}[\widetilde{\mathbb{O}}])$.

Y is affine over G/P satisfies the situation of the previous remark.

$G/U \to G/P$ is an L-principal bundle, $T^*(G/U) \times X_L \to G/U$ is affine.

We can consider all of these quantizations, Hamiltonian reductions as sheaves over the smooth variety G/P.

There is a sheaf of $\mathfrak{z}$-algebra $\mathcal{D}_{\mathfrak{z}}$ over G/P as the universal quantization of Y. This is also the quantization of $Y_{\mathfrak{z}} \to Y$.

41. Harish-Chandra Bimodules

2026-06-08

$\mathfrak{g}$ be a Lie algebra, $\mathcal{B}$ be a $U(\mathfrak{g})$-bimodule. A good filtration on $\mathcal{B}$ is a filtration:

$$0 = \mathcal{B}_{-1} \subseteq \mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \cdots \subseteq \mathcal{B}_n \subseteq \cdots,$$

such that

- $\bigcup_{i=0}^\infty \mathcal{B}_i = \mathcal{B}$;
- $[U_i(\mathfrak{g}), \mathcal{B}_j] \subseteq \mathcal{B}_{i+j-1}$, $U_i(\mathfrak{g}) \cdot \mathcal{B}_j \subseteq \mathcal{B}_{i+j}$;
- $\mathrm{gr}(\mathcal{B})$ is finitely generated as $S(\mathfrak{g})$-module.

Theorem 41.1. *$\mathcal{B}$ has a good filtration if and only if $\mathcal{B}$ is finitely-generated (left or right $U(\mathfrak{g})$-module) and adjoint action of $\mathfrak{g}$ on $\mathcal{B}$ is locally finite.*

Proof. $\square$

Proposition 41.2. *If $\mathcal{B}$ is locally finite under the adjoint action of $\mathfrak{g}$, then $\mathcal{B}$ is left f.g. $\Leftrightarrow$ bi-f.g. $\Leftrightarrow$ right f.g.*

Proof. $\square$

42. Principal Connections and Picard Groups

2026-06-29

X be a smooth manifold, $\mathcal{V}$ a vector bundle on X, it's associated frame bundle is the principal bundle $\mathcal{M}$. Note that $\mathcal{V} = \mathcal{M} \times^{\mathrm{GL}_n} \mathbb{C}^n$.

A connection on $\mathcal{V}$ is a map $\nabla : \Gamma(X, TX) \to \Gamma(X, \mathfrak{gl}(\mathcal{V}))$. While a connection on $\mathcal{M}$ is a GL-equivariant bundle map $\theta : \mathcal{E}_M \to \mathfrak{gl}(V)_M$, split the Atiyah sequence:

$$0 \to \mathfrak{gl}(V)_M \to \mathcal{E}_M \to TX \to 0.$$

Restriction of θ to TX gives a connection on $\mathcal{V}$, this is a bijection.

Here we compute some Picard groups of symplectic singularities, and their Poisson resolutions.

$\widetilde{\mathbb{O}}$ be a nilpotent cover of a complex semisimple Lie group G, $\widetilde{\mathbb{O}_L}$ be a nilpotent cover of a Levi subgroup L. And there is a Poisson resolution

$$\pi : Y = G \times^P (X_L \times \mathfrak{p}^\perp) \to X.$$

Proposition 42.1.
- $\mathrm{Pic}(X) = 0$;
- $\mathrm{Pic}(X^{reg}) = X^*(\widetilde{H})$, *where* $\widetilde{H}$ *is the stabilizer in* $\widetilde{G}$ *of a point in* $\widetilde{\mathbb{O}}$.

Proof. First assertion is due to X is conical.

For the second assertion, since $\widetilde{\mathbb{O}} \subseteq X^{reg} \subseteq X$ and $\mathrm{codim}_X(X \backslash \widetilde{\mathbb{O}}) \geq 2$, we have

$$\mathrm{Pic}(X^{reg}) = \mathrm{Pic}(\widetilde{\mathbb{O}}).$$

Since $\widetilde{\mathbb{O}}$ is a homogeneous space of $\widetilde{G}$, $\widetilde{\mathbb{O}} \simeq \widetilde{G}/\widetilde{H}$ ($\widetilde{G}$ is the simply-connected cover of G),

$$\mathrm{Pic}(\widetilde{\mathbb{O}}) = \mathrm{Pic}(\widetilde{G}/\widetilde{H}) = \mathrm{Pic}^{\widetilde{G}}(\widetilde{G}/\widetilde{H}) = X^*(\widetilde{H}).$$

The second equality is due to the fact that $\widetilde{G}$ is simply-connected, so any line bundle on $\widetilde{G}/\widetilde{H}$ admits a $\widetilde{G}$-equivariant structure. And $\mathcal{O}_X$ has only one $\widetilde{G}$-equivariant structure, since $H^1(G; \mathcal{O}_X^*(X)) = 0$. □

Proposition 42.2. *There is a short exact sequence*

$$0 \to \mathrm{Pic}(Y) \to \mathrm{Pic}(Y^{reg}) \to \mathrm{Pic}(X_L^{reg}) \to 0.$$

Proof. Take the open Bruhat cell $U^- \subseteq G/P$.

Then

$$\pi^{-1}(U^-) \simeq U^- \times (X_L \times \mathfrak{p}^\perp).$$

U^- and $\mathfrak{p}^\perp$ are affine spaces, so $\mathrm{Cl}(\pi^{-1}(U^-)) = \mathrm{Cl}(X_L) = \mathrm{Pic}(X_L^{reg})$.

The map $\mathrm{Pic}(Y^{reg}) \to \mathrm{Pic}(X_L^{reg})$ is just the restriction map $\mathrm{Cl}(Y) \to \mathrm{Cl}(\pi^{-1}(U^-))$, and the kernel is $\mathrm{Cl}(Y \backslash \pi^{-1}(U^-))$.

$$\mathrm{Pic}(Y) \xrightarrow{div} \mathrm{Cl}(Y) \xrightarrow{res} \mathrm{Cl}(\pi^{-1}(U^-)),$$

the composition can also realized as

$$\mathrm{Pic}(Y) \xrightarrow{res} \mathrm{Pic}(\pi^{-1}(U^-)) \xrightarrow{div} \mathrm{Cl}(\pi^{-1}(U^-)),$$

while $\mathrm{Pic}(\pi^{-1}(U^-)) = 0$ the composition is zero.

On the other hand, if a divisor D on Y is mapped to zero in $\mathrm{Cl}(\pi^{-1}(U^-))$, then D is linearly equivalent to a divisor supported on $Y \backslash \pi^{-1}(U^-)$, it is linearly equivalent to $\sum_i n_i \pi^{-1}(D_i)$ (suppose that $(G/P) \backslash U^- = \cup_i D_i$, where D_i are divisors.), and $\pi^{-1}(D_i) = \mathrm{div}(\pi^*(\mathcal{O}(D_i)))$ they are Cartier divisors, so D is Cartier. This shows that the kernel of $\mathrm{Cl}(Y) \to \mathrm{Cl}(\pi^{-1}(U^-))$ is $\mathrm{Pic}(Y)$. □

Proposition 42.3. $\mathrm{Pic}(Y) = \mathrm{Pic}(G/P) = X^*(\widetilde{L})$.

Proof. Then $\pi^* : \mathrm{Pic}(G/P) \to \mathrm{Pic}(Y)$ is injective, since π has a section $G/P \hookrightarrow Y$. For any line bundle $\mathcal{L}$ on Y the proof of previous proposition shows that $\mathcal{L}$ is isomorphic to some $\mathcal{O}(\sum_i \pi^{-1}(D_i)) = \pi^*(\mathcal{O}(\sum_i D_i))$. $\square$

Theorem 42.4. *If $\widetilde{\mathbb{O}_L}$ is birationally rigid, then $H^2(Y^{reg}, \mathbb{C}) = \mathrm{Pic}(Y) \otimes_{\mathbb{Z}} \mathbb{C} = \mathfrak{z}(\mathfrak{l})$.*

43. Formal quantization and Star product

2026-07-25

For a Poisson variety X a formal quantization of X is a pair $(\mathcal{D}_\hbar, \theta)$ where $\mathcal{D}_\hbar$ is sheaf of $\mathbb{C}[[h]]$-algebra flat over $\mathbb{C}[[\hbar]]$ and complete in $\hbar$-adic topology, and $\theta : \mathcal{D}_\hbar/(\hbar) \xrightarrow{\sim} \mathcal{O}_X$ is a Poisson isomorphism. Isomorphism classes of formal quantizations of X is denoted by $Quant(X)$

When $X = \mathrm{Spec}(A)$ is affine, use micro-localization, we can identify the formal quantization of X with a formal quantization of A.

A star product on A is a $\mathbb{C}[[\hbar]]$ bi-linear map

$$\star : A[[\hbar]] \times A[[\hbar]] \to A[[\hbar]], \quad f \star g = \sum_{i=0}^{\infty} D_i(f,g)\hbar^i,$$

where $D_i : A \times A \to A$ are bi-differential operators, such that

- $D_0(f,g) = fg$;
- $D_1(f,g) - D_1(g,f) = \{f,g\}$;
- $\star$ is associative, 1 is the unit.

Two star products $\star$ and $\star'$ are called equivalent if there is a $\mathbb{C}[[\hbar]]$-linear automorphism $T : A[[\hbar]] \to A[[\hbar]]$,

$$T = \mathrm{id} + \sum_{i=0}^{\infty} T_i \hbar^i, \quad T_i : A \to A,$$

such that $T(f \star g) = T(f) \star' T(g)$.

Theorem 43.1. *There is a bijection:*

$$Quant(X) \xrightarrow{\sim} \{\textit{Star products on } A\}/\textit{equivalence}.$$

Proof. Step 1: For any formal quantization $A_\hbar$ of A, there exist a $\mathbb{C}[[\hbar]]$-linear map $\Phi : A[[\hbar]] \xrightarrow{\sim} A_\hbar$.

By definition of formal quantization, we have an isomorphism $\theta : A_\hbar/(\hbar) \simeq A$. We can choose a $\mathbb{C}$-linear section $s : A \to A_\hbar$ of θ, then extend it to a $\mathbb{C}[[\hbar]]$-linear map $\Phi : A[[\hbar]] \to A_\hbar, \quad \sum a_i\hbar^i \mapsto \sum s(a_i)\hbar^i$. This map is surjective, for any $a \in A_\hbar$, let $a_0 = \theta(a) \in A$, then $a - s(a_0) \in \hbar A_\hbar$, continue this process, we can find a_i for each $i \in \mathbb{N}$ such that $a = \sum s(a_i)\hbar^i$. This map is injective, if $\sum s(a_i)\hbar^i = 0$, then $s(a_0) \in \hbar A_\hbar$, so $a_0 = \theta \circ s(a_0) = 0$, continue this process, we can find $a_i = 0$ for each $i \in \mathbb{N}$.

So we can transport the product on $A_\hbar$ to $A[[\hbar]]$, which satisfies the conditions of star product without the bi-differential condition. The choice of Φ is not unique, if we choose another Φ', then $\Phi' \circ \Phi^{-1}$ is a $\mathbb{C}[[\hbar]]$-linear automorphism of $A[[\hbar]]$, which is of the form $T = \mathrm{id} + \sum_{i=0}^{\infty} T_i\hbar^i$, and T gives an equivalence between the two transported products.

Step 2: There exist a choice of Φ such that the transported product on $A[[\hbar]]$ satisfies the bi-differential condition.
We need the fact that $HH^i(A) \simeq HH^i_{\text{diff}}(A)$.

□

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